

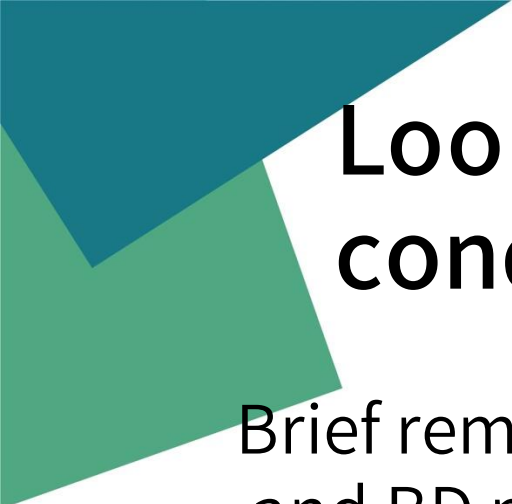
Observing plastic evolution related to high-field conditioning

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Looking for the missing conditioning – structure link

Brief reminder - link between sub-surface plastic evolution
and BD nucleation mechanism

Some info on trials :

Soft versus hard

STEM+FIB versus the world

Results:

Current state and something to think about

Non linear evolution of moving dislocations leads to critical transition

- Large number of sources and sinks for dislocations is available (Initially an assumption - I'll show later)

Governing equations:

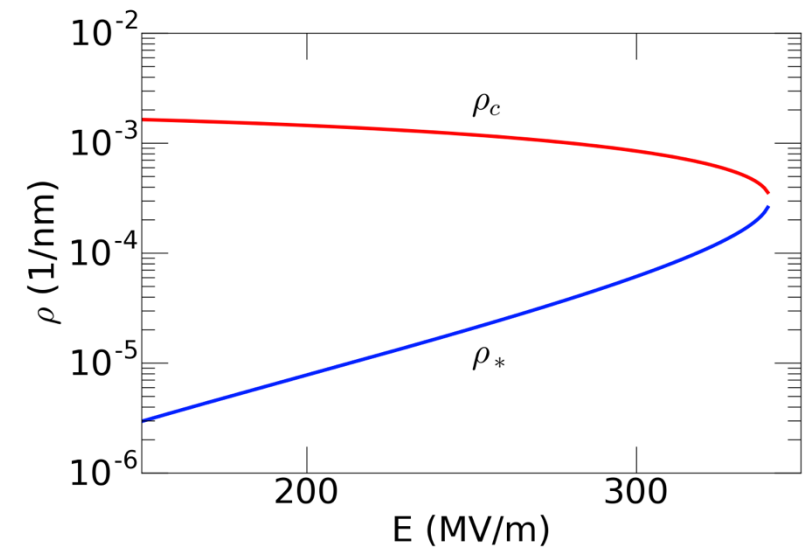
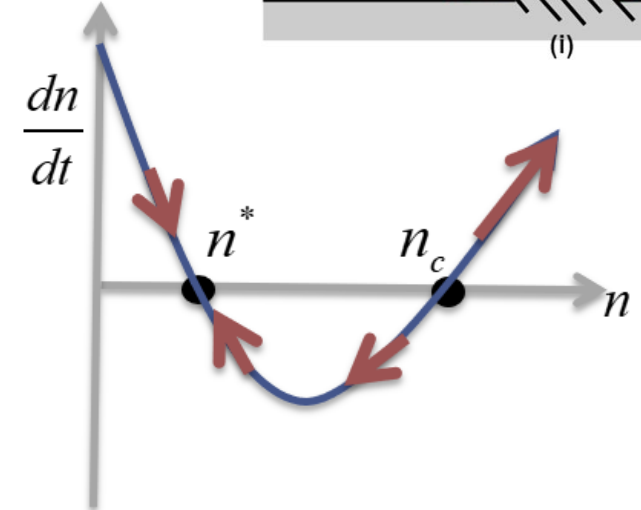
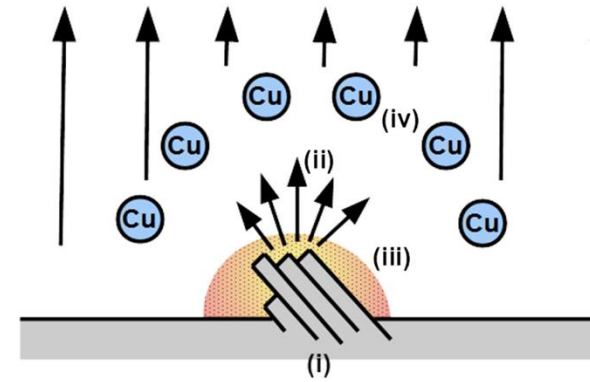
- Increase in moving population – interactions with field and moving dislocations.
- Arrest due to collisions
- Cooperative critical transition in moving dislocation population generates nucleation event.

This transition: a runaway in the moving dislocations - can lead to a nucleation event through its effects on the surface state*

*effect leading to a jump in current emission

PHYSICAL REVIEW LETTERS **120**, 124801 (2018)

Stochastic Model of Breakdown Nucleation under Intense Electric Fields



Model results vs observations - Rate variation with field

Consistent fit to breakdown rate dependence on field
Importance of temperature variation.

- "Classical" scenario: Temperature effect on BDR versus field curves.
Dynamic - Ramping up field at various rates.
Average "field for BD"

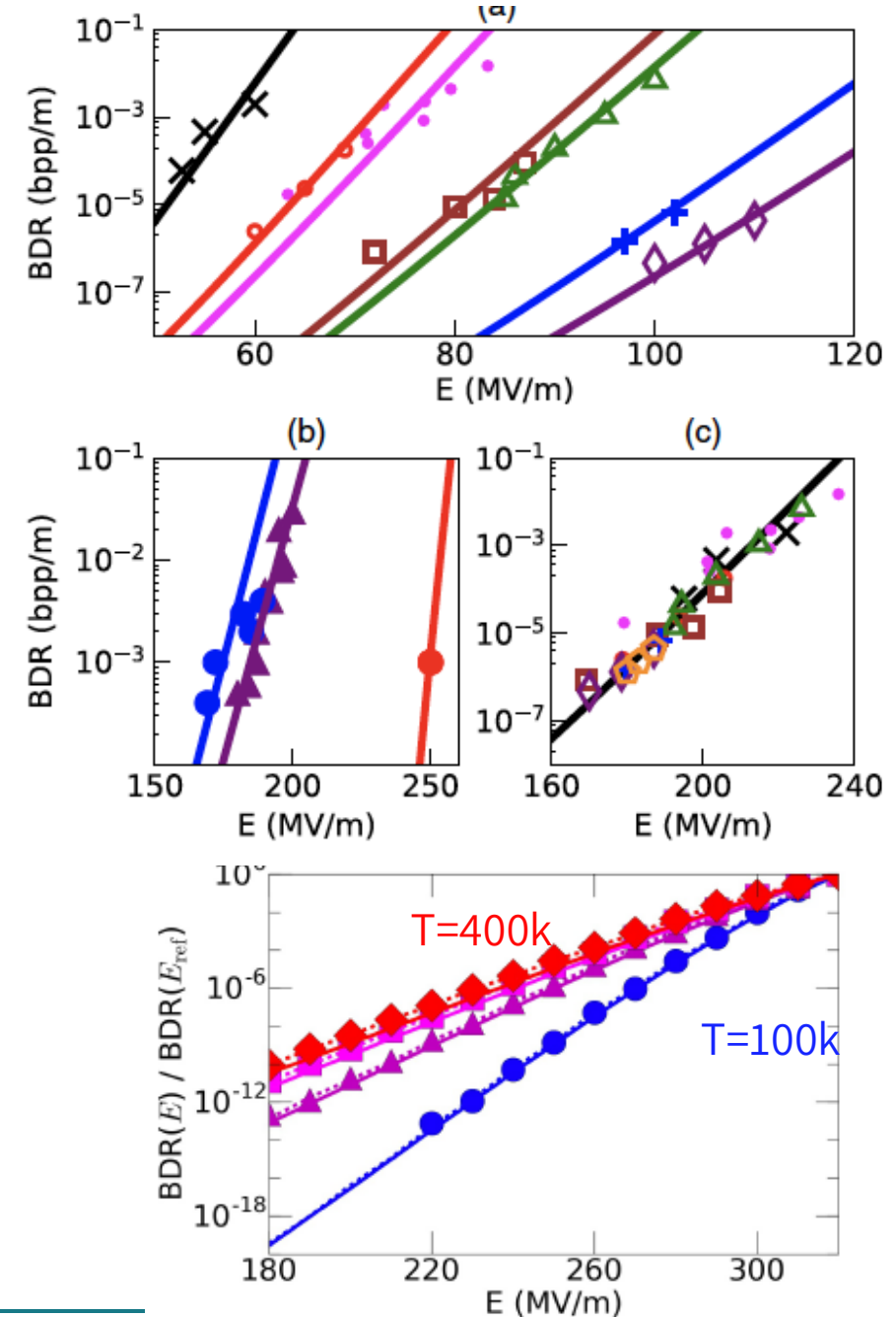
Time dependency - Non-Linear regime.

Prior to BD (as the field is increased), stronger fluctuations in the population leading to fluctuations in the dark current should be observed!

PHYSICAL REVIEW ACCELERATORS AND BEAMS **22**, 083501 (2019)

Editors' Suggestion

Theory of electric field breakdown nucleation due to mobile dislocations

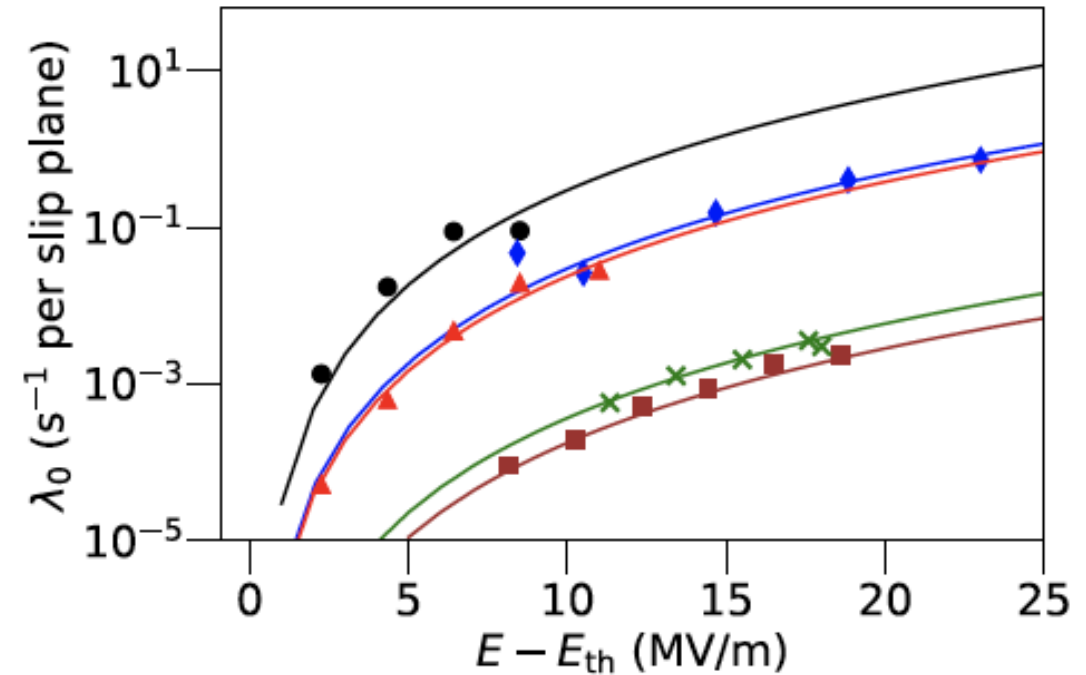
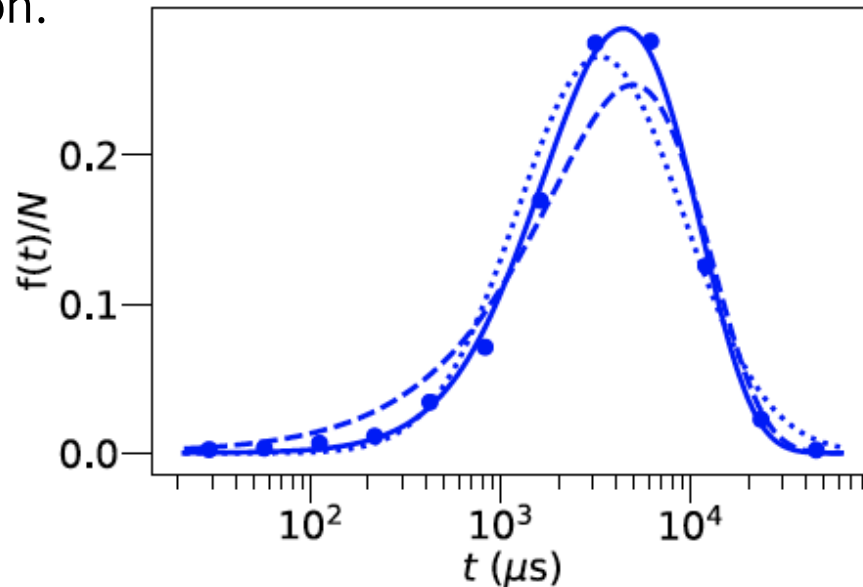


Fluctuations - offer a possible explanation to spikes in dark currents (prior to BD)

- Event rate variation with the field is reproduced by MDDF based equation.

$$\lambda_0 = \frac{25\kappa C_t c}{G^2 b \Delta \rho} \sigma_0^2 \exp\left(-\frac{E_a - \Omega \sigma_0}{k_B T}\right)$$

- The distribution fits the expected hypoexponential distribution.



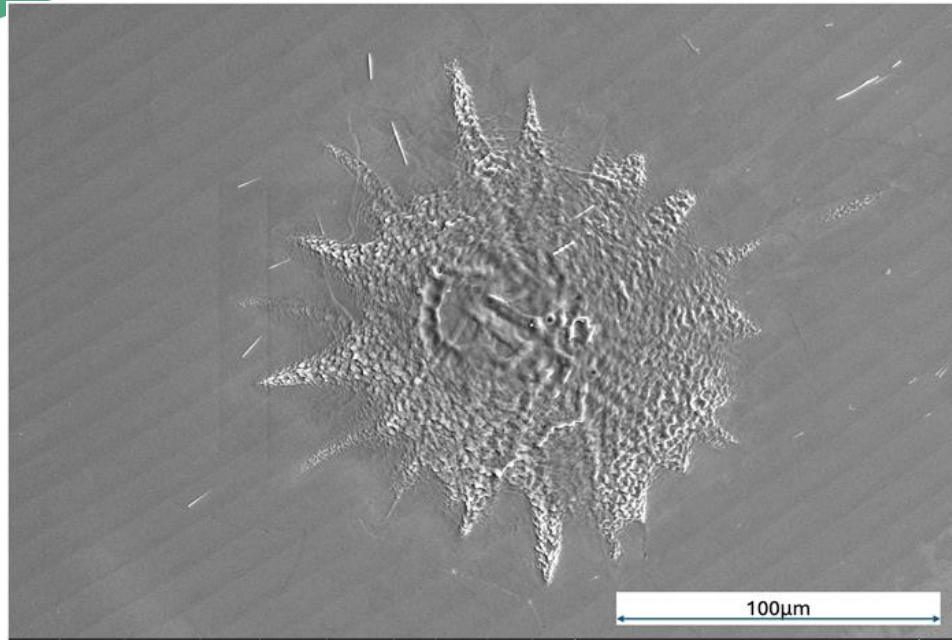
PHYSICAL REVIEW ACCELERATORS AND BEAMS **23**, 123501 (2020)

Dark current spikes as an indicator of mobile dislocation dynamics under intense dc electric fields

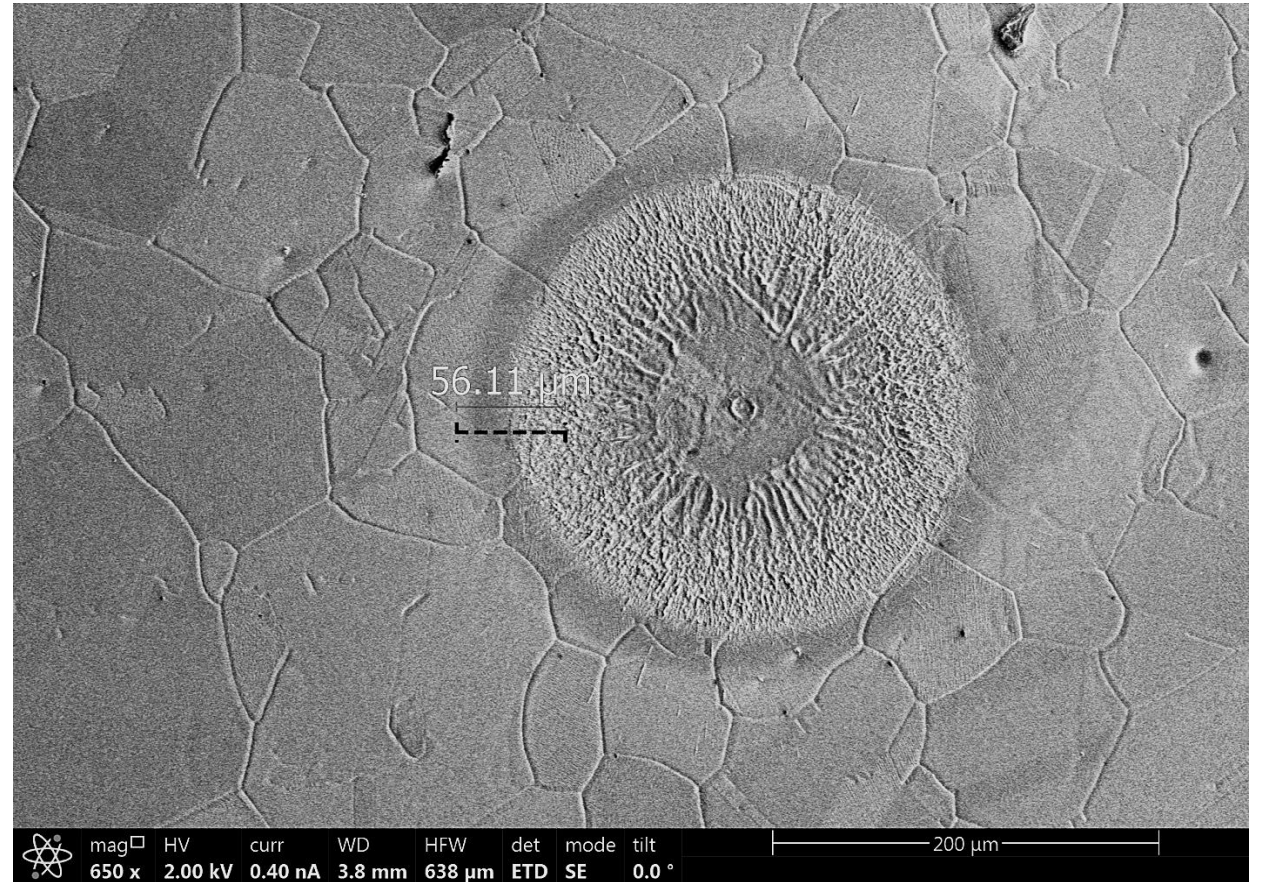
Eliyahu Zvi Engelberg¹, Jan Paszkiewicz², Ruth Peacock², Sagy Lachmann,¹
Yinon Ashkenazy,^{1,*} and Walter Wuensch²

Cu Deposition allows insights to plasma evolution

SEM of post-BD events: Insights into BD evolution.

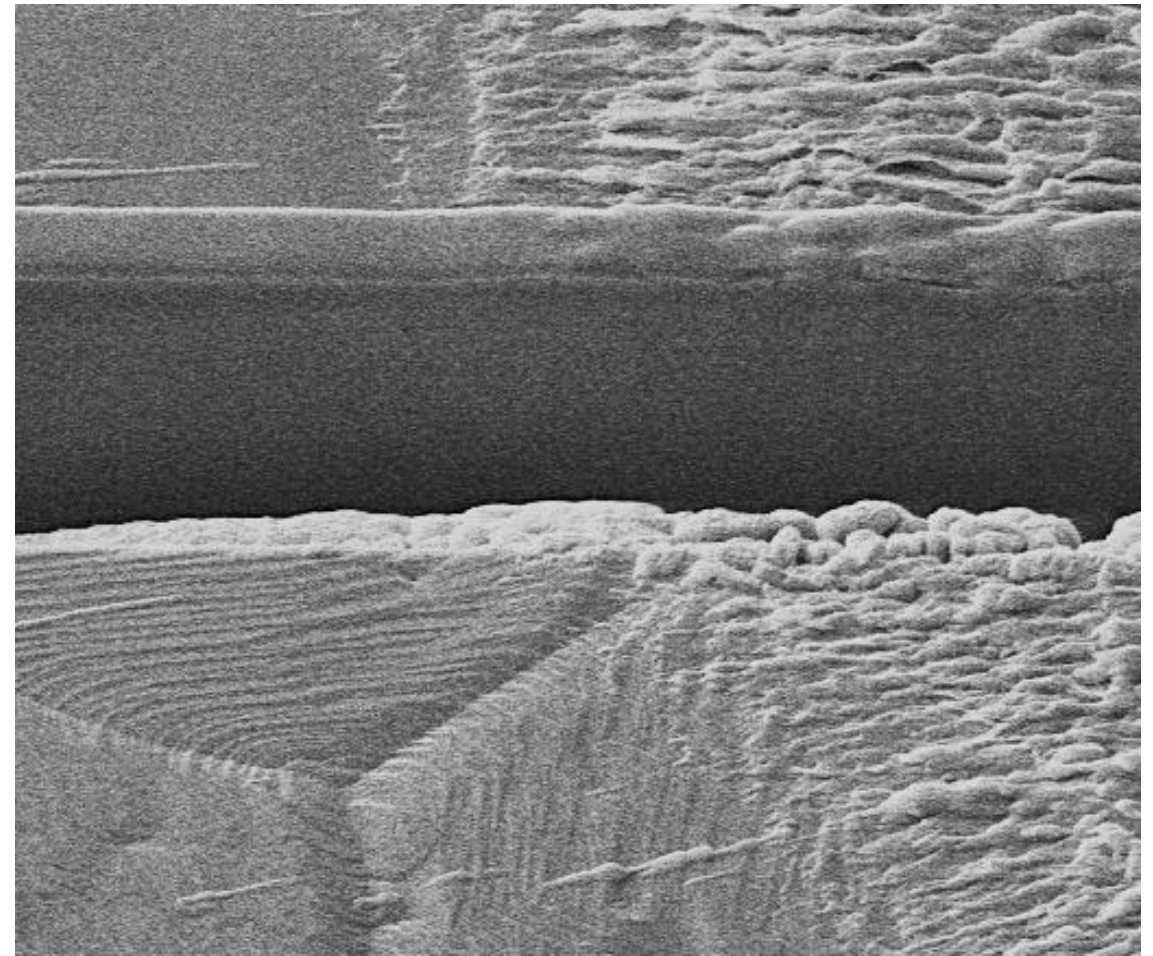
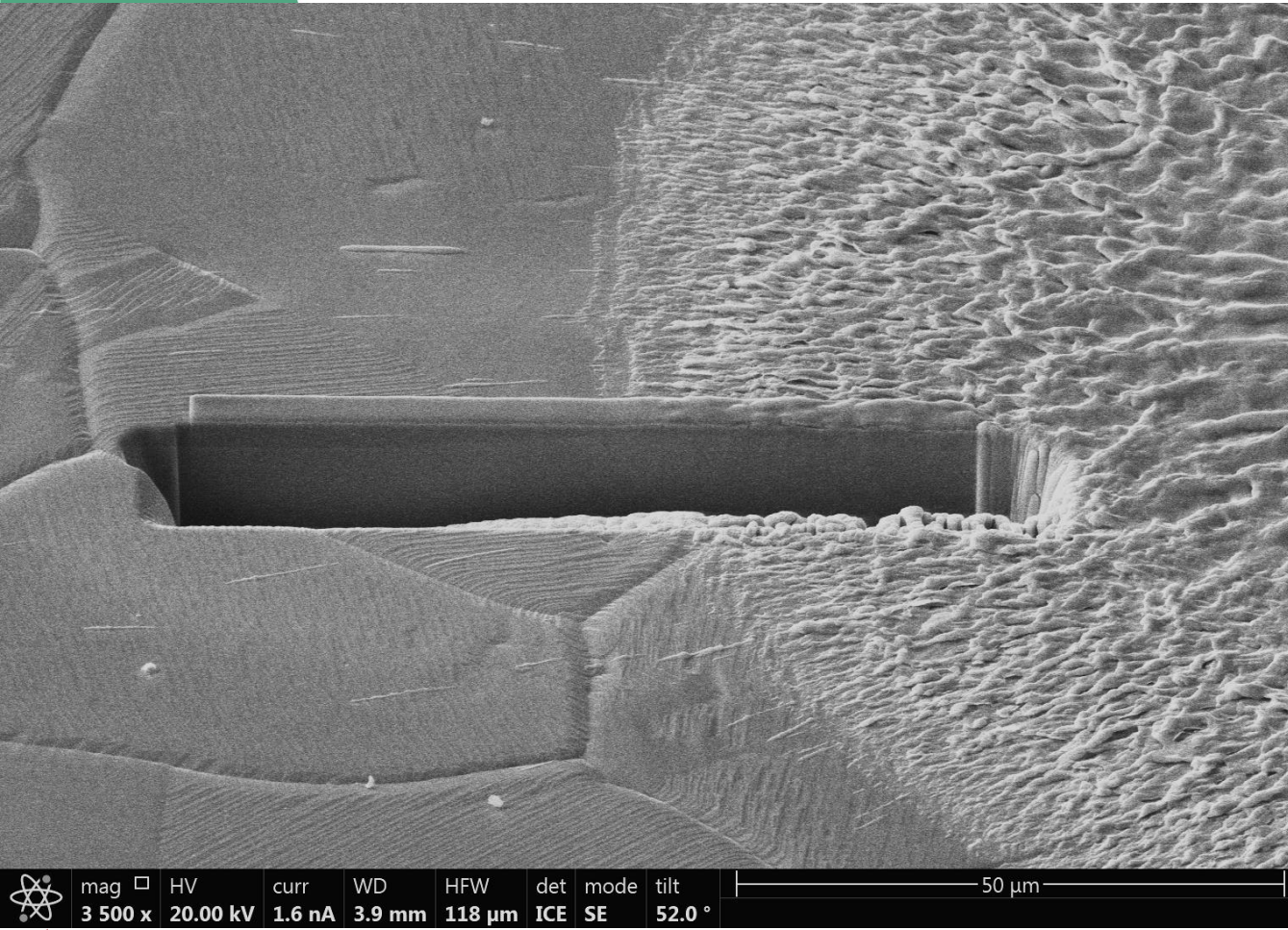


SEM of a star-like BD site on the cathode (30K).
Outside the deposition region - machining grooves are visible

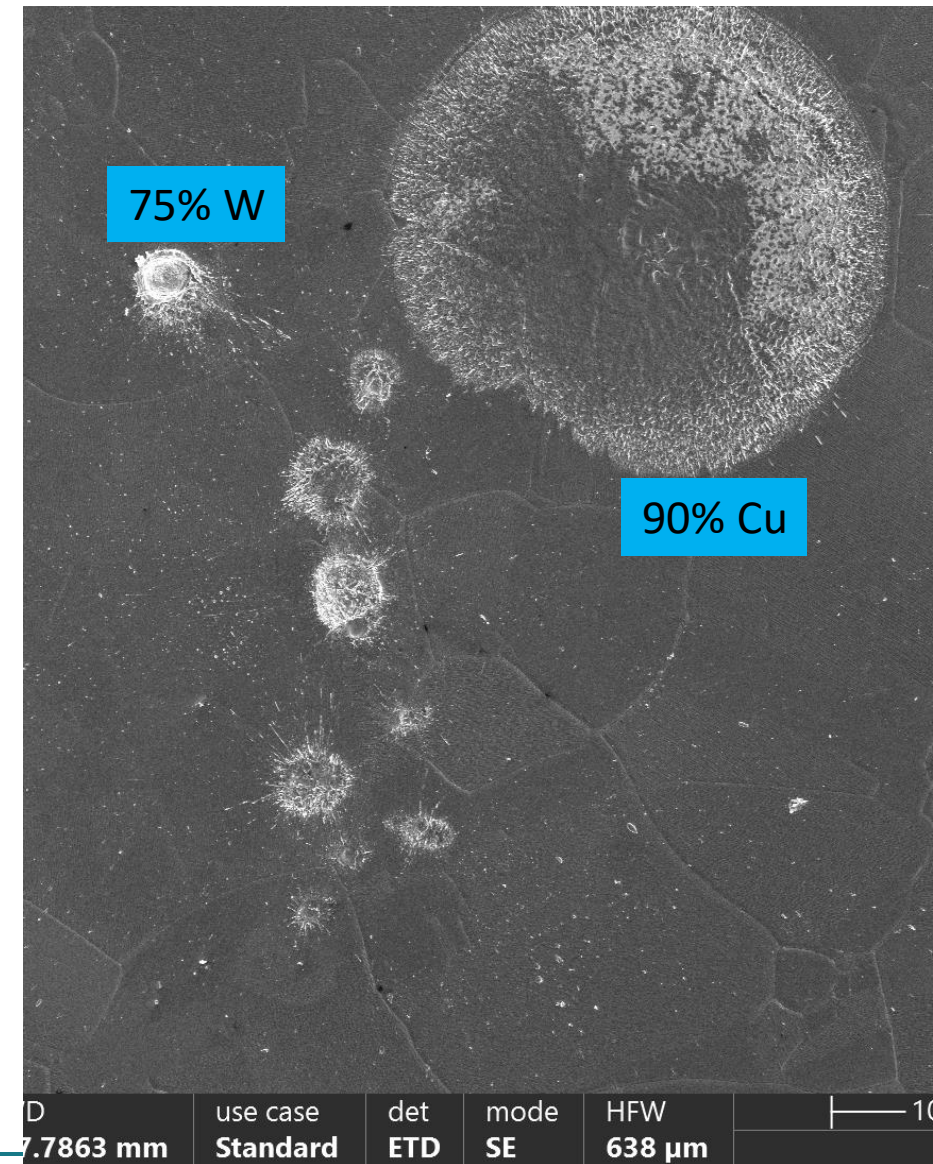
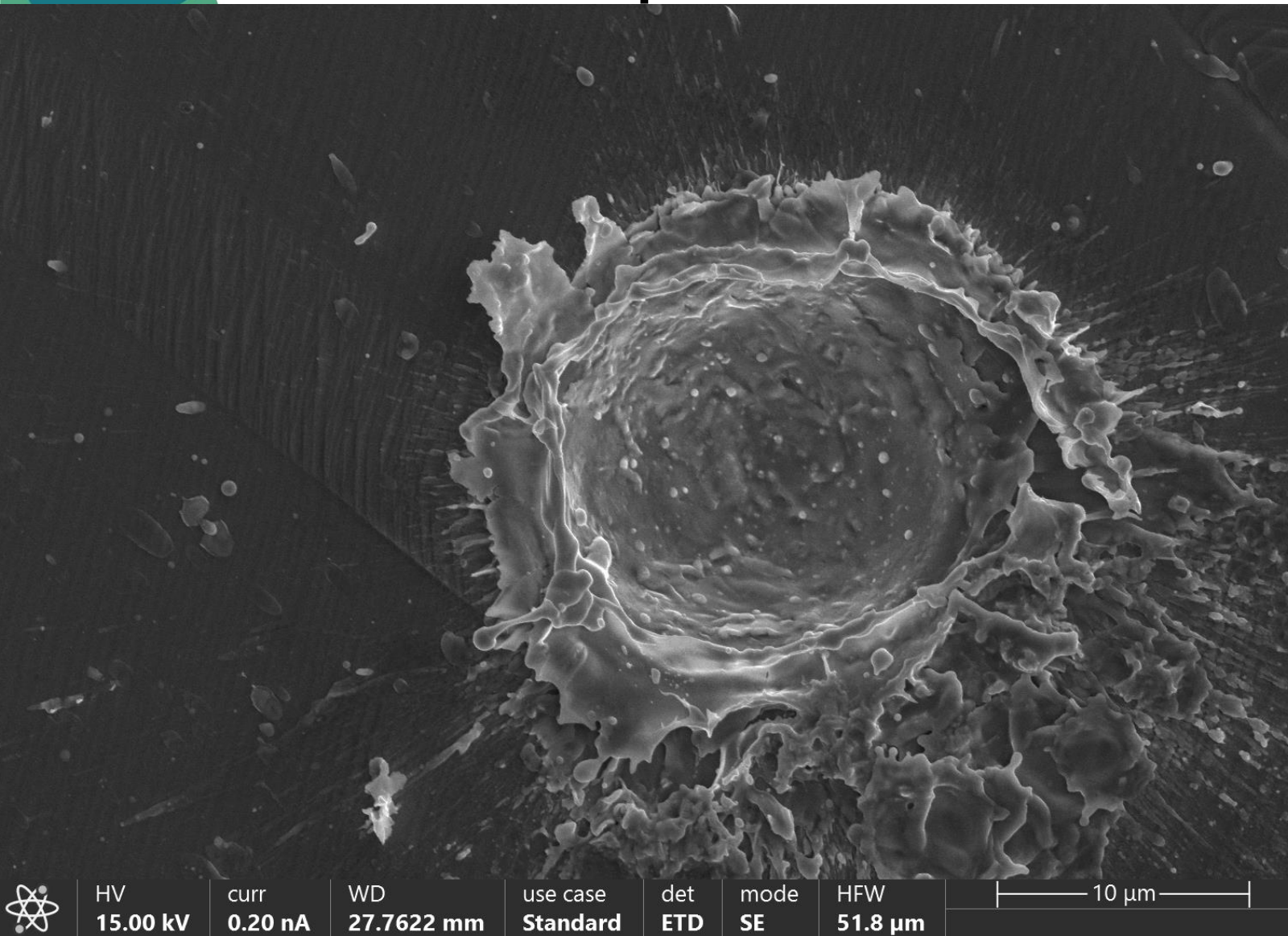


J. Appl. Phys. 137, 193302 (2025)
<https://doi.org/10.1063/5.0258896>

Cu Deposition allows insights into plasma evolution – more to be done here



Different plasmas – different damage



75% W

90% Cu

What do we expect to see below the surface?

- Transition is due to an avalanche of moving dislocations.
- Before the critical transition, dislocations move, but at low numbers.
- No quasi-static evolution of surface features.

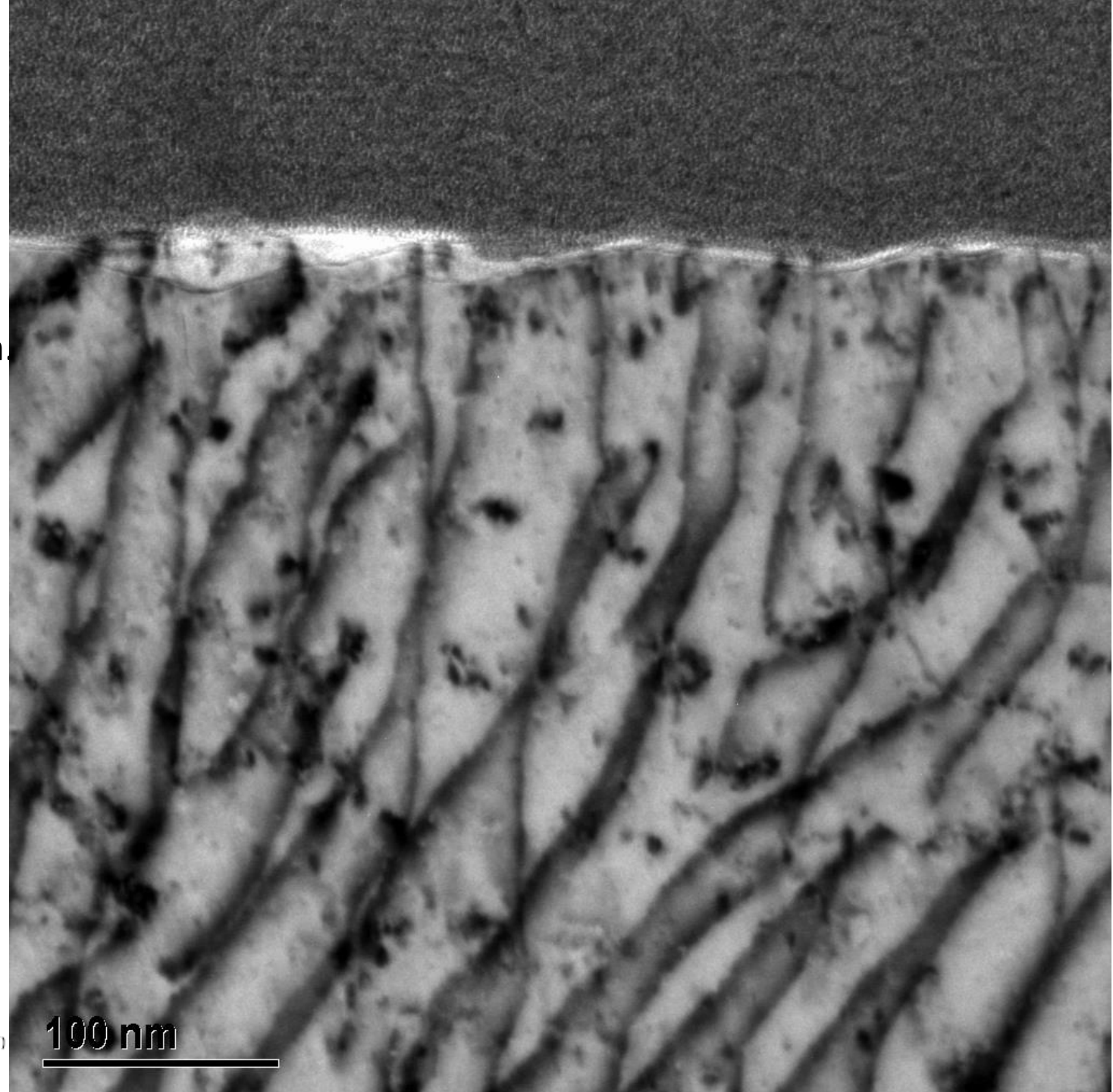
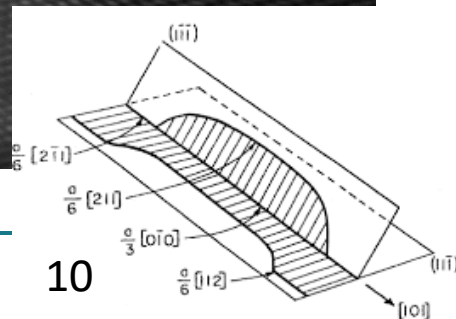
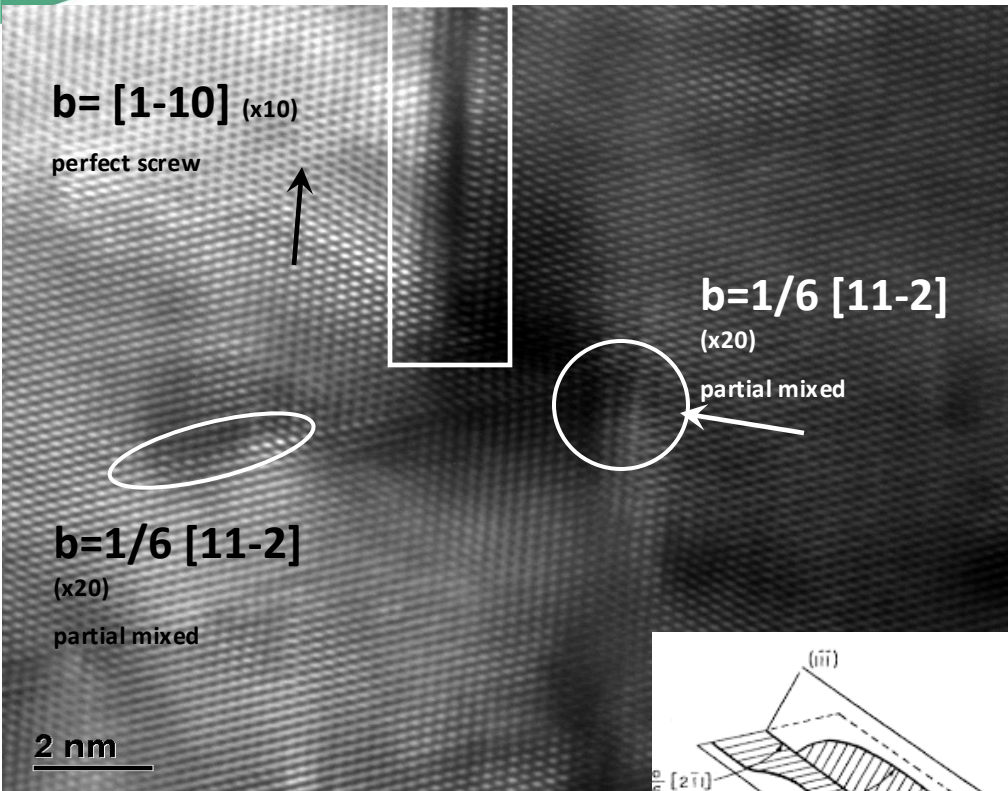
If true, then:

- Conditioning due to dislocation motion affecting the sub-surface region in terms relevant to dislocation motion and interaction.
- The mechanism depends on surface interaction, dictating a preferred direction. So, the effect is limited to the top sub-micron region.
- We should be able to observe a large density of mobile dislocation sources.
- **We should see the results of exhausting easy sources.**

FIB+TEM

Observe an Intricate dislocation pattern below the surface that correlates well with the surface structure.

Sadly a result of sample preparation.



FIB damage of Cu and possible consequences for miniaturized mechanical tests

D. Kiener^{a,b,*}, C. Motz^b, M. Rester^b, M. Jenko^c, G. Dehm^{b,d}

D. Kiener et al. / Materials Science and Engineering A 459 (2007) 262–272

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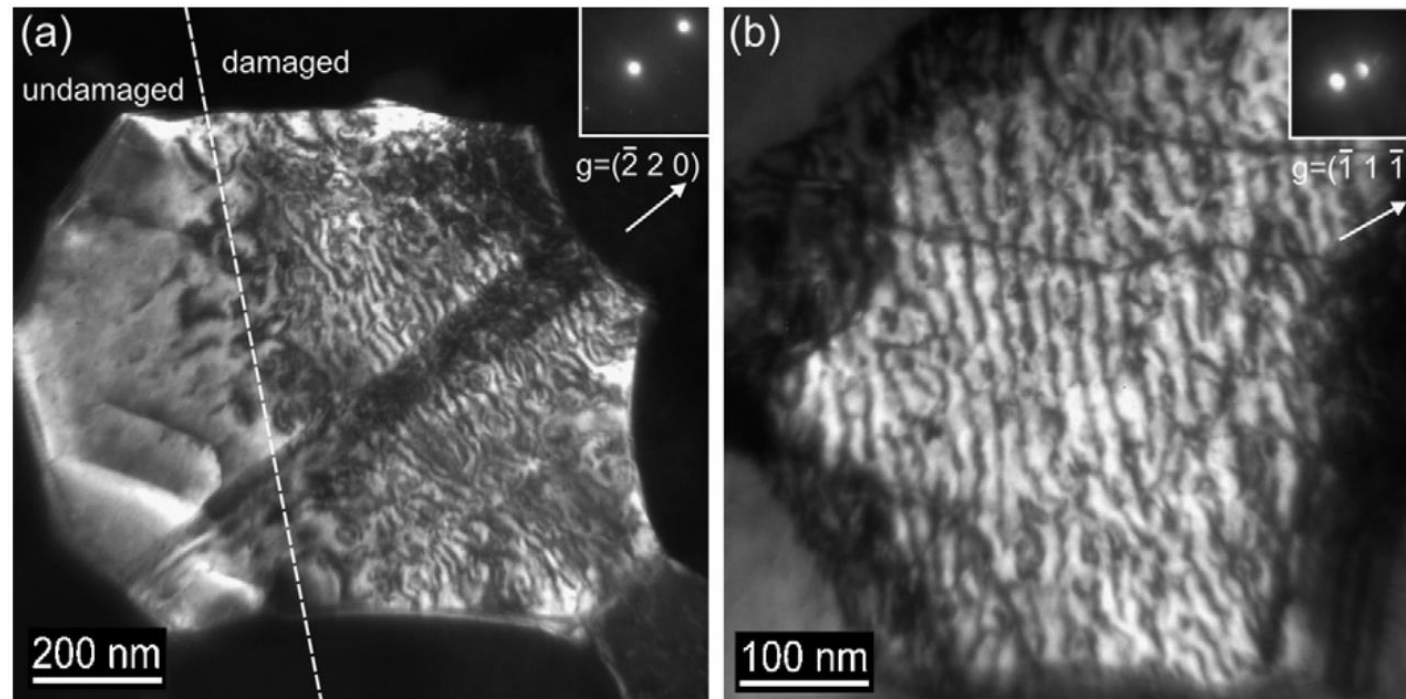


Fig. 2. (a) Dark-field TEM image of a Cu grain situated at the border of an ion damaged and undamaged region. The right part of the grain was exposed to Ga^+ ions with a kinetic energy of 30 keV and an ion current of 50 pA. The g_{-220} diffraction condition is shown in the inset. (b) Bright-field TEM micrograph of the damage pattern of a grain bombarded by 5 keV Ga^+ ions and an ion current of 100 pA. The g_{-11-1} diffraction condition is shown in the inset.

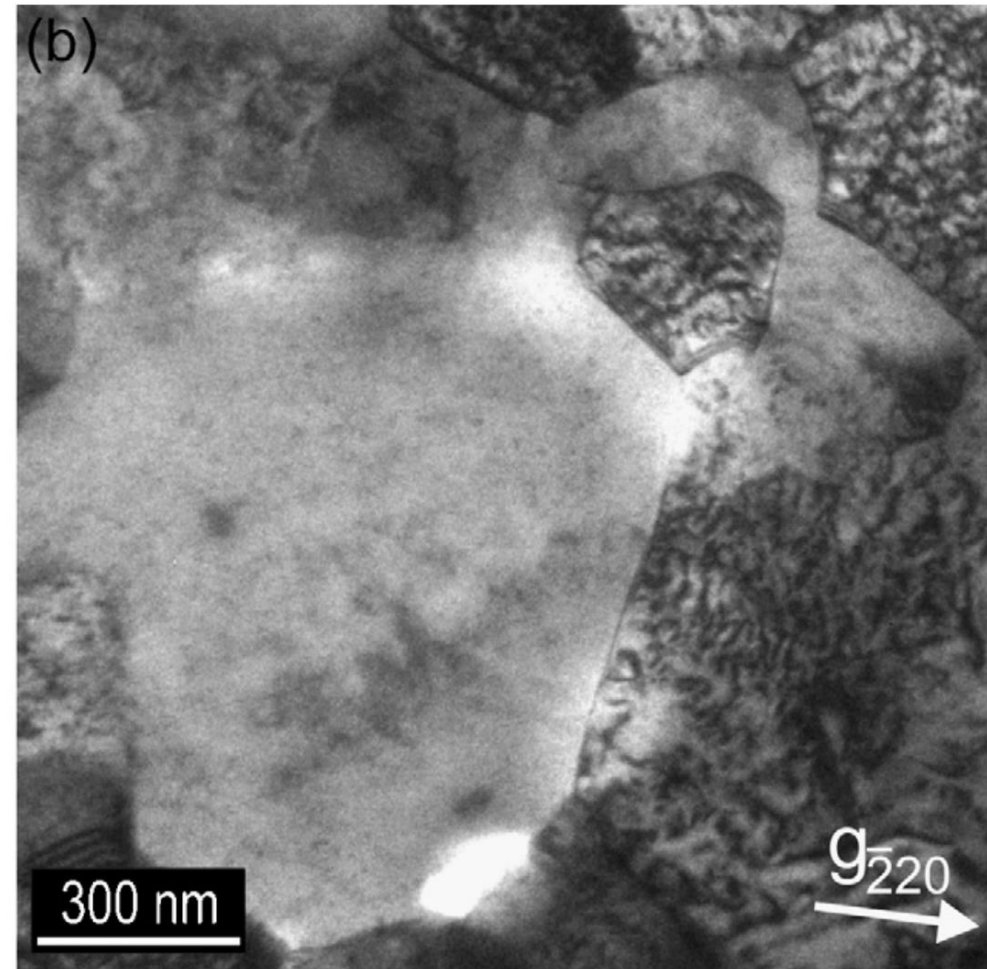
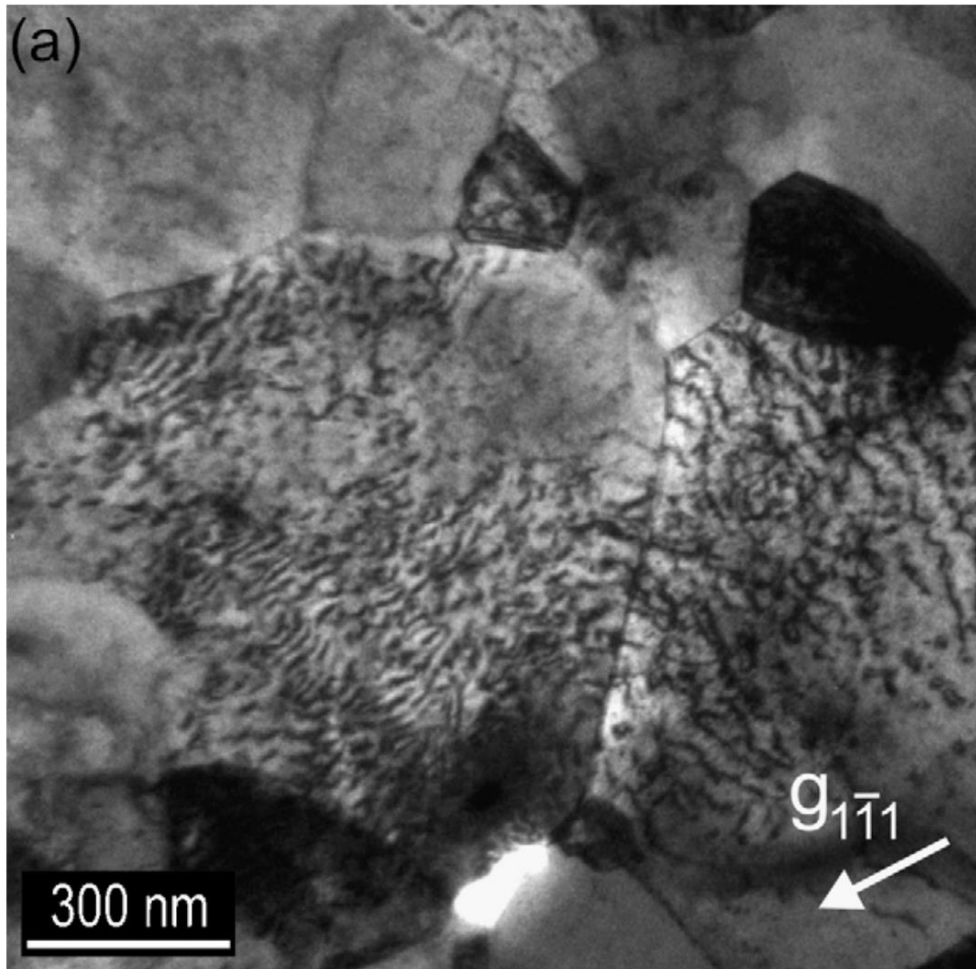


Fig. 3. Bright-field TEM images of Cu grains exposed to perpendicular 30 keV Ga^+ ions and an ion current of 50 pA under different diffraction conditions. The diffraction vector is shown for the grain on the right hand side. The strain contrast vanishes for certain conditions, as can be seen for several grains (compare a and b).

- Perpendicular, high voltage, high current – lead to FIB generated dislocation structures in all grains (observable in different diffraction directions)

So what can we do

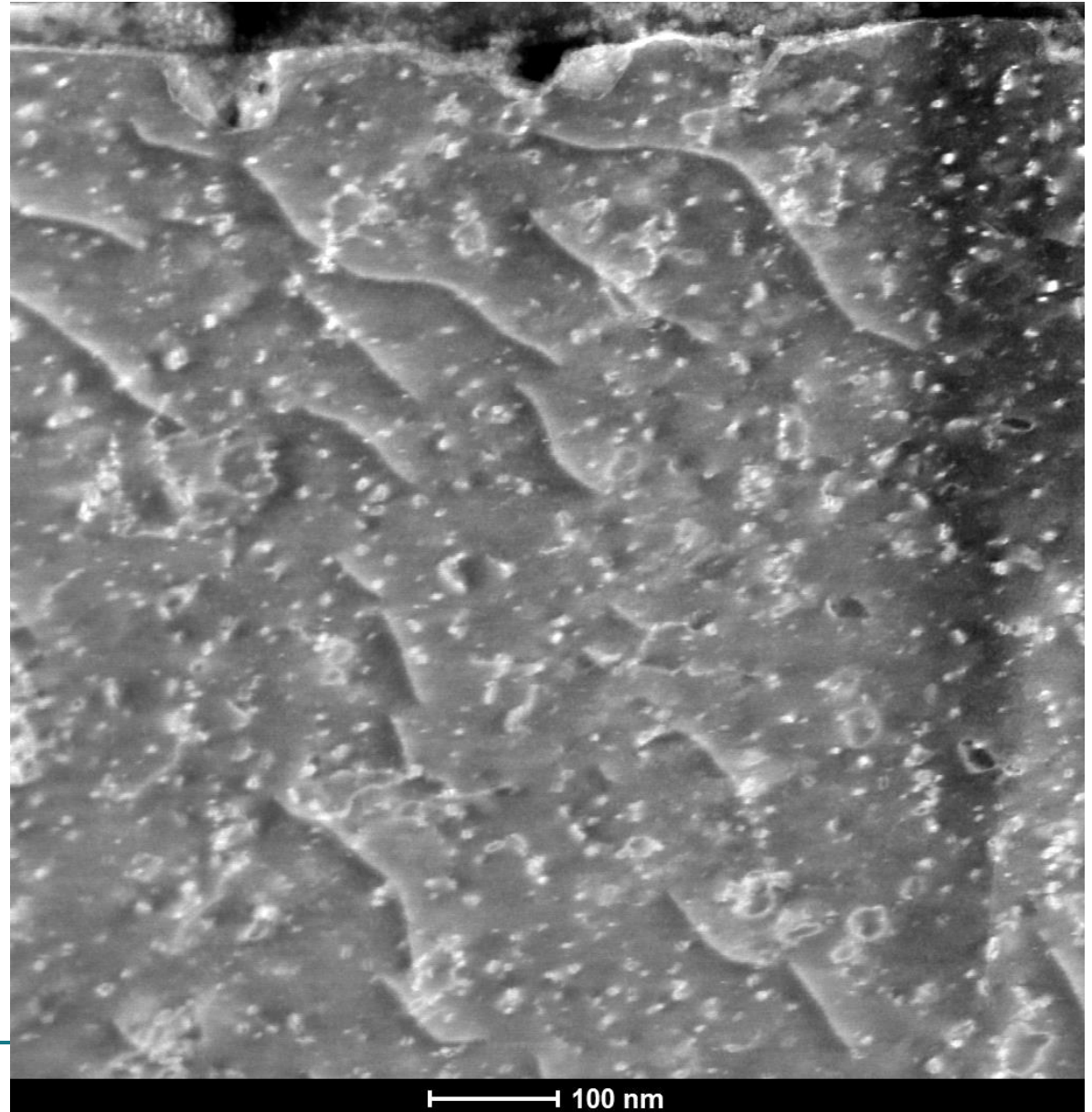
4. Discussion

In order to reduce the damage introduced by the Ga^+ ion bombardment, several possibilities exist. For example 👍 use of a protective layer, 👍 low incidence angle of the impinging Ga^+ ions, 👍 low ion energies, and 👍 optimized milling geometries in order to avoid redeposition. A further possibility to reduce the

Completely covered? Almost... but not enough.
Still in large grains of Cu – FIB damage prevent observing small scale structures

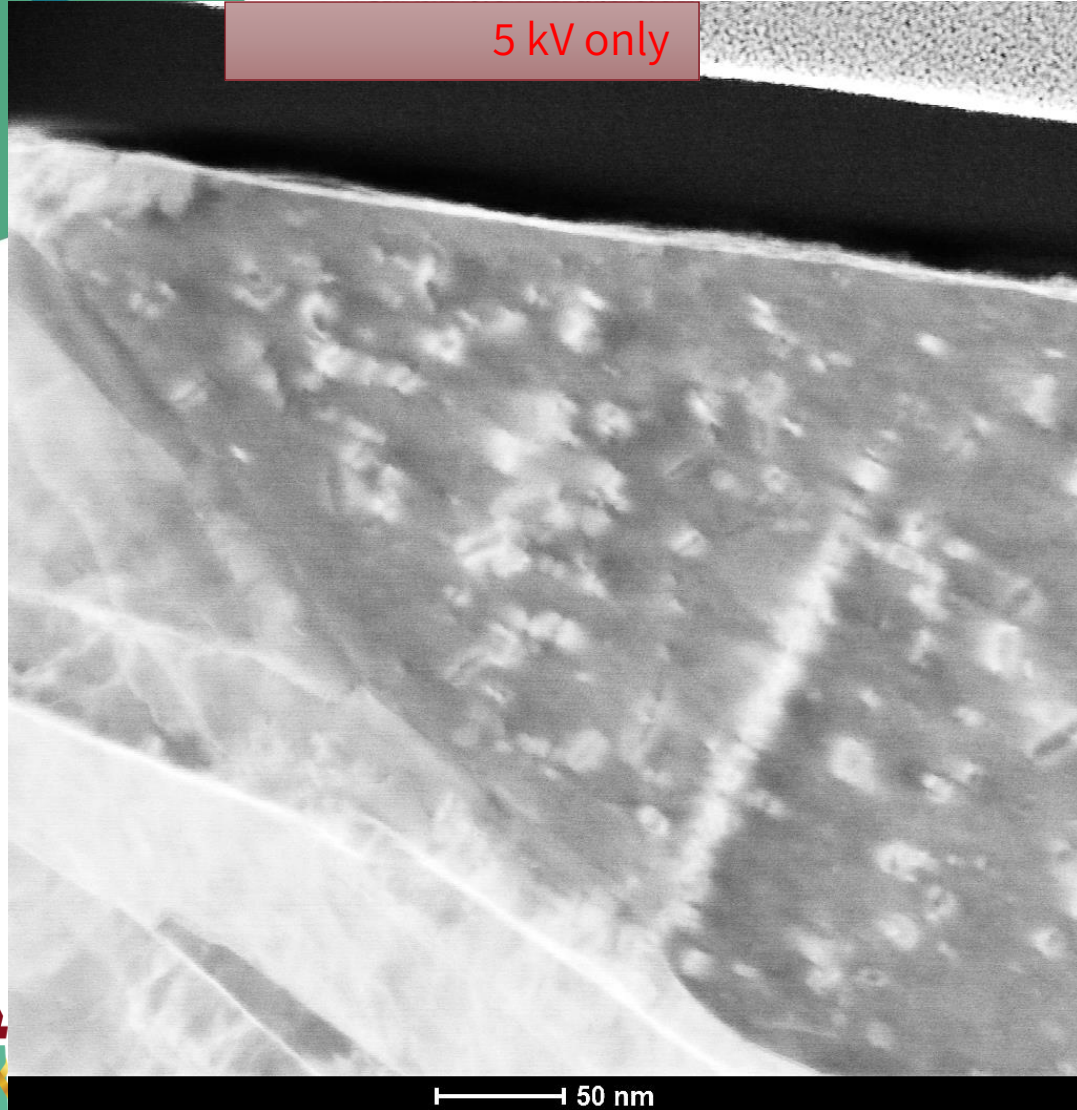
Soft Cu is hard to look at

After taking all precautions, soft Cu continues to exhibit damage from FIB sample production, including dislocations and loops.

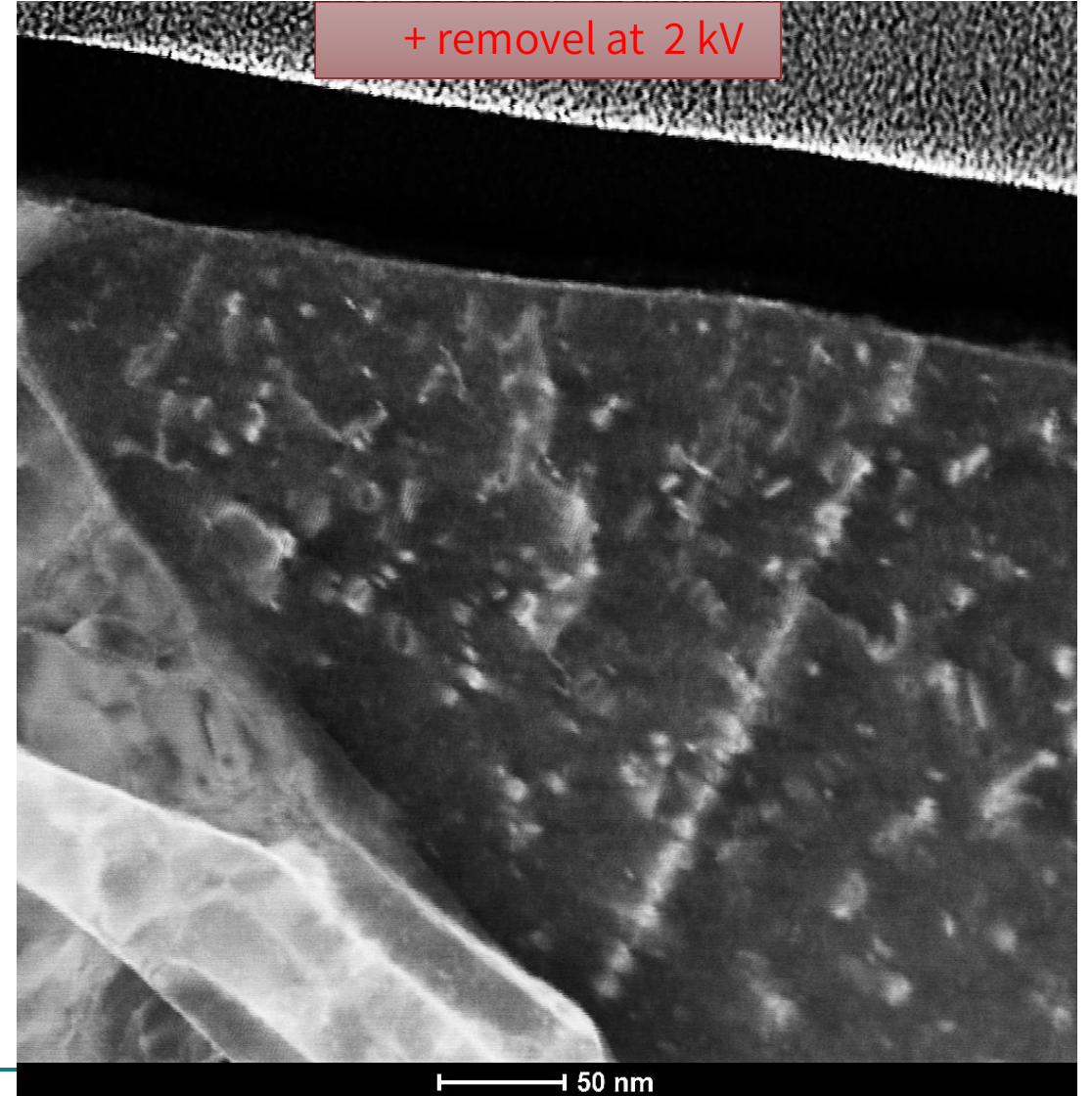


Even hard Cu shows a high concentration of artifacts

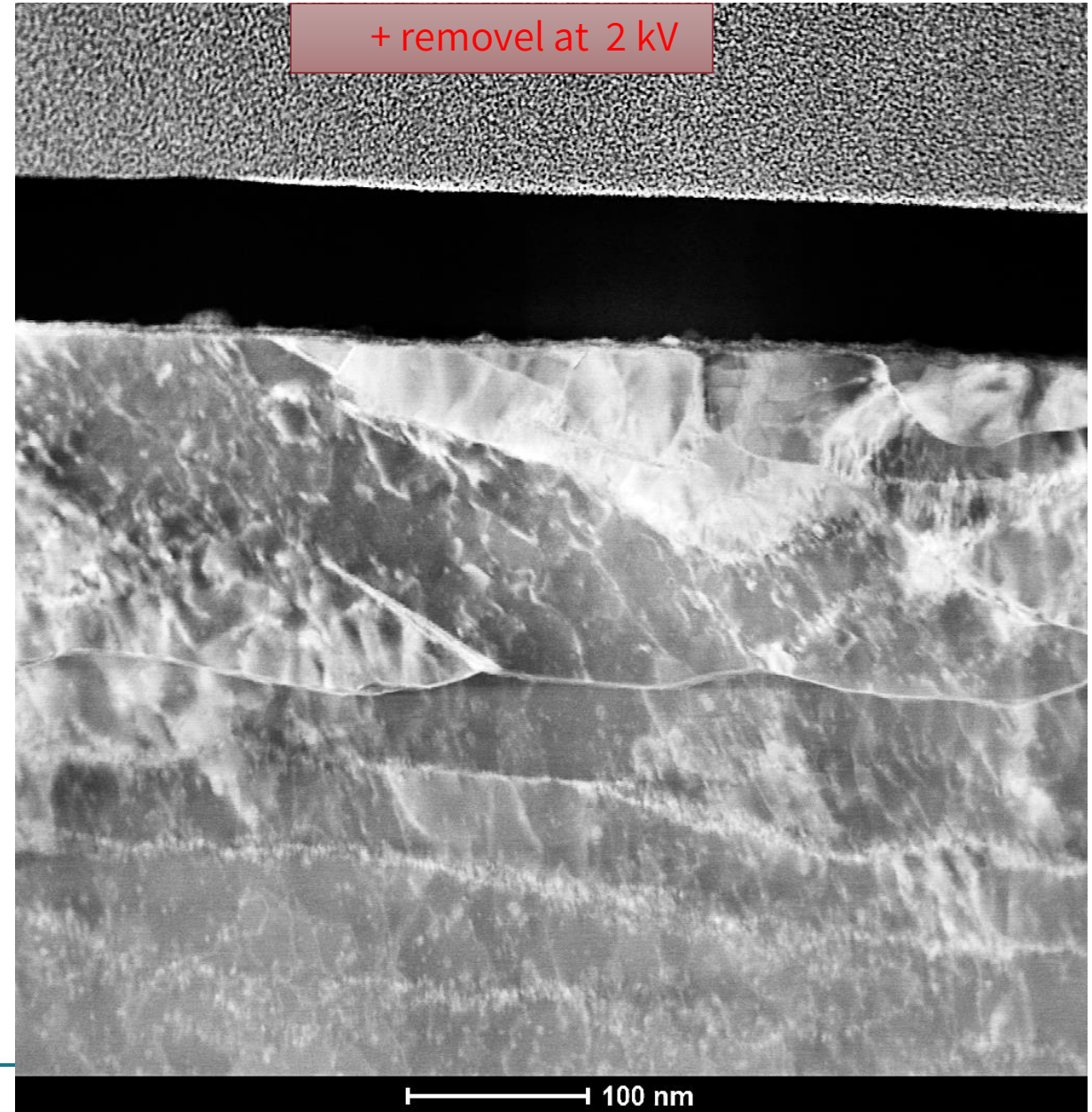
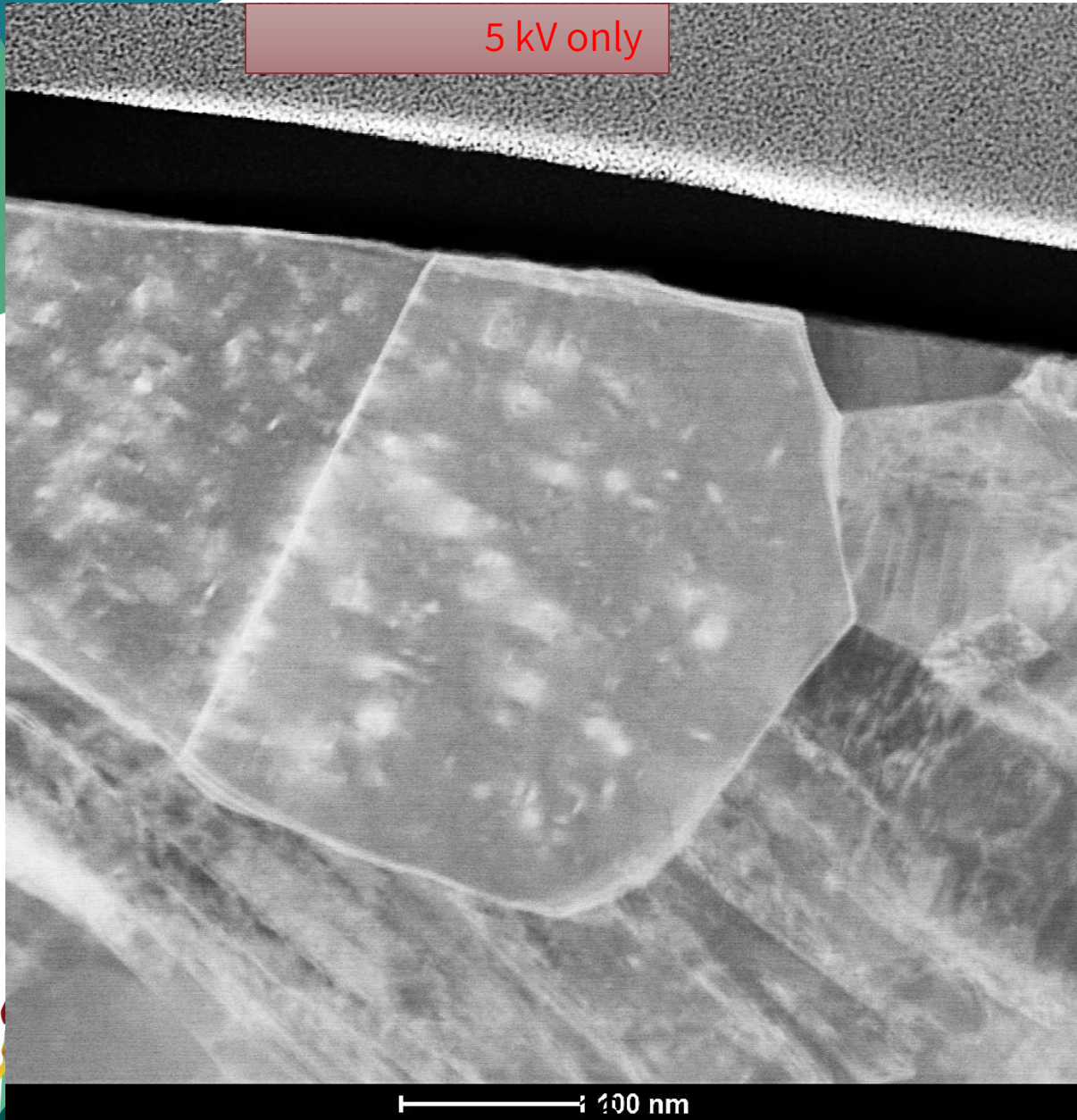
5 kV only



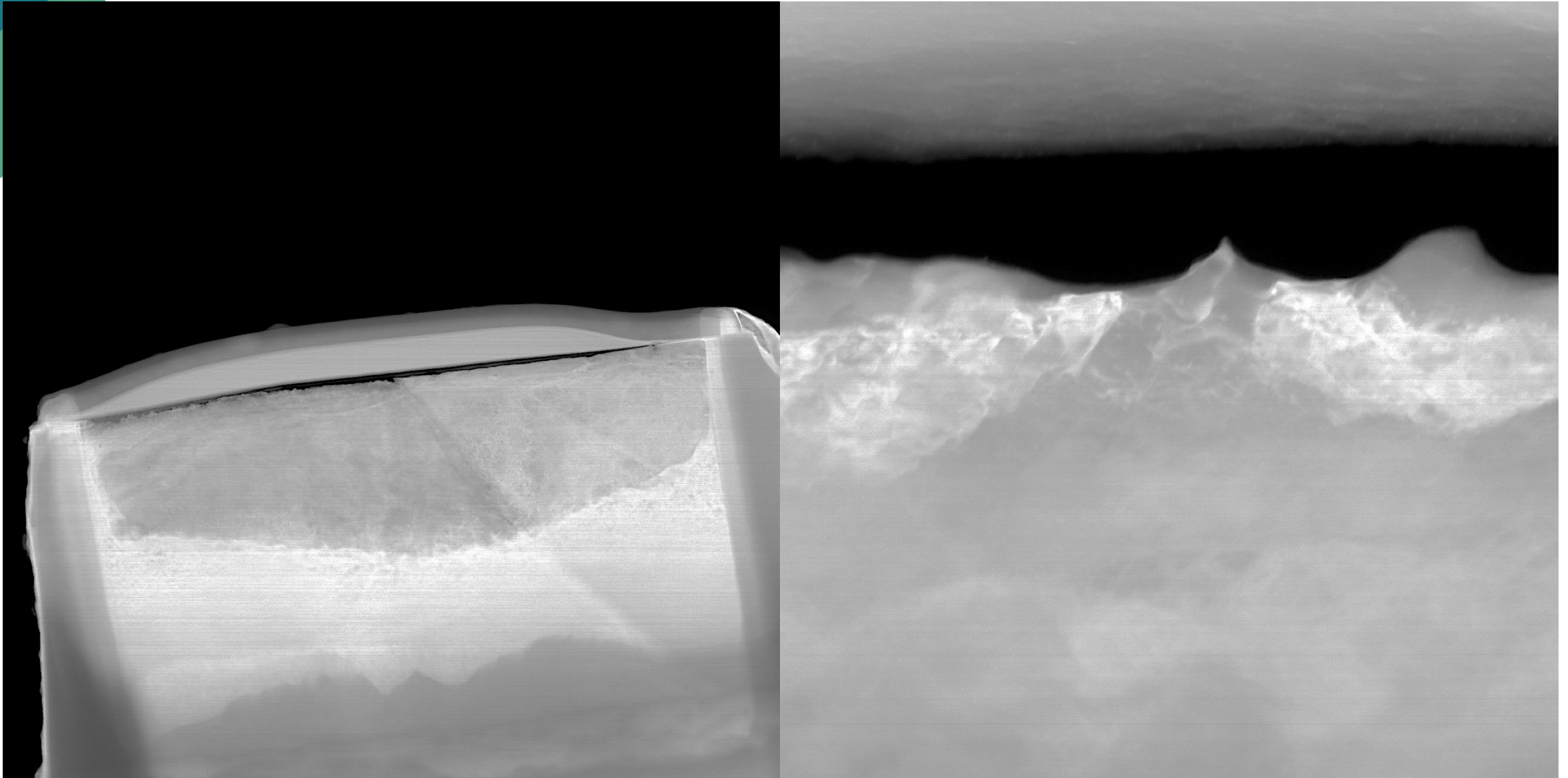
+ removal at 2 kV



But- large features are not affected



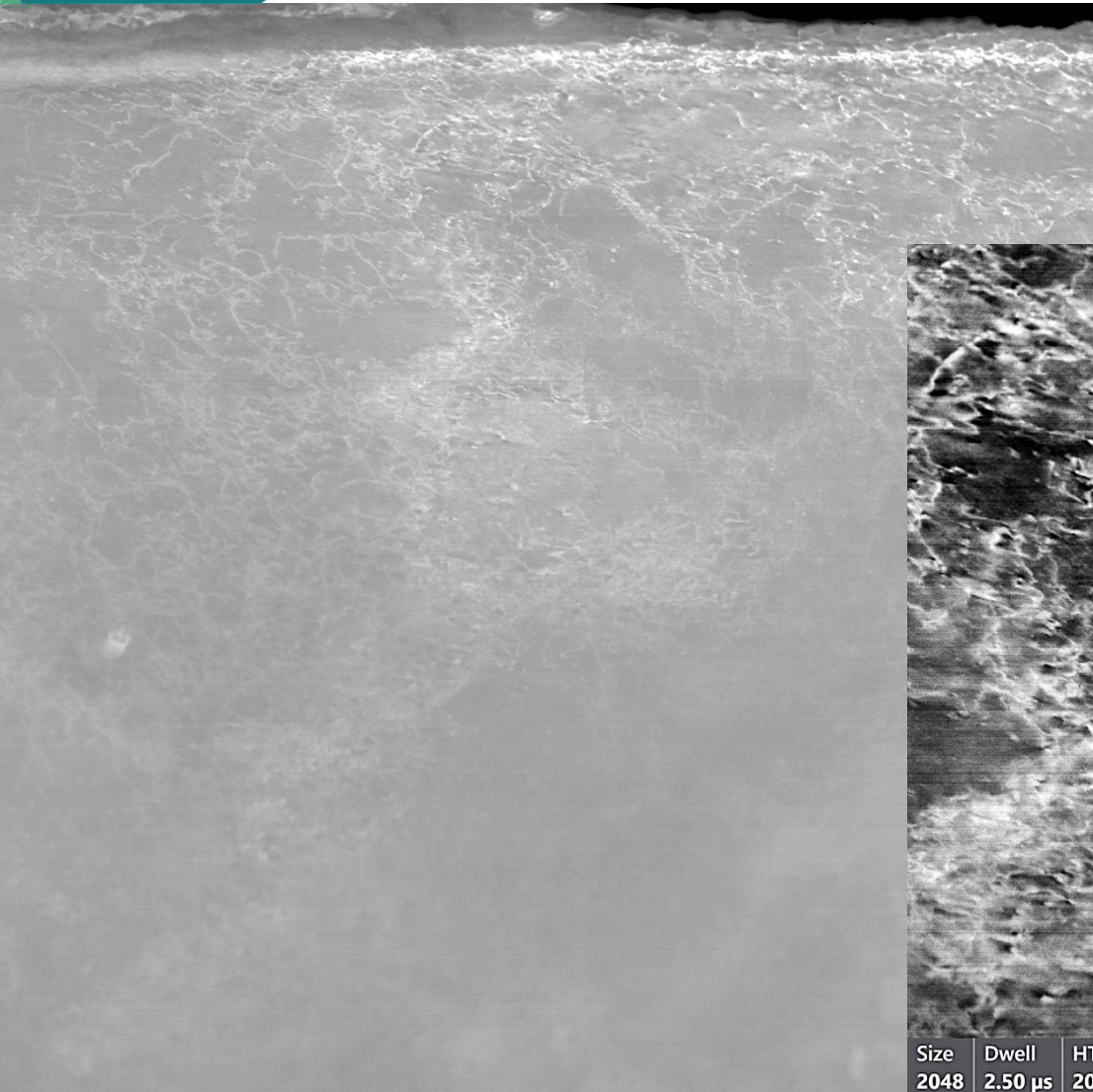
Electropolishing - no radiation damage, but surface is lost



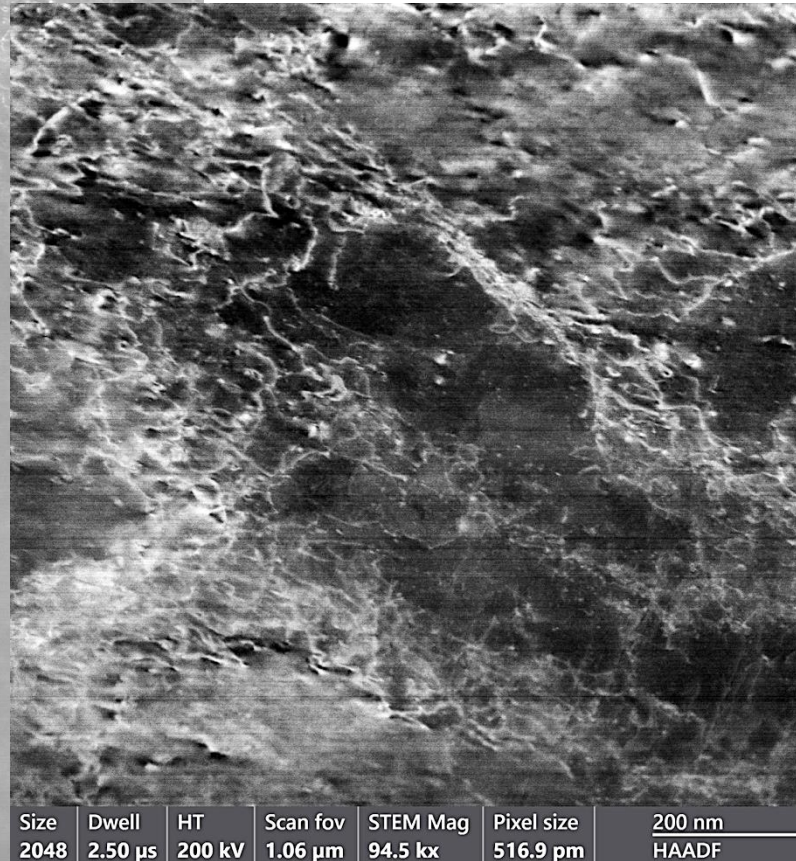
Size	Dwell	HT	Scan fov	STEM Mag	Pixel size	CL	<u>2 μm</u>	Size	Dwell	HT	Scan fov	STEM Mag	Pixel size	Spot	<u>50 nm</u>
2048	2.50 μ s	200 kV	17 μ m	5.87 kx	8.313 nm	98 mm	HAADF	2048	2.50 μ s	200 kV	374 nm	268 kx	182.4 pm	3	HAADF



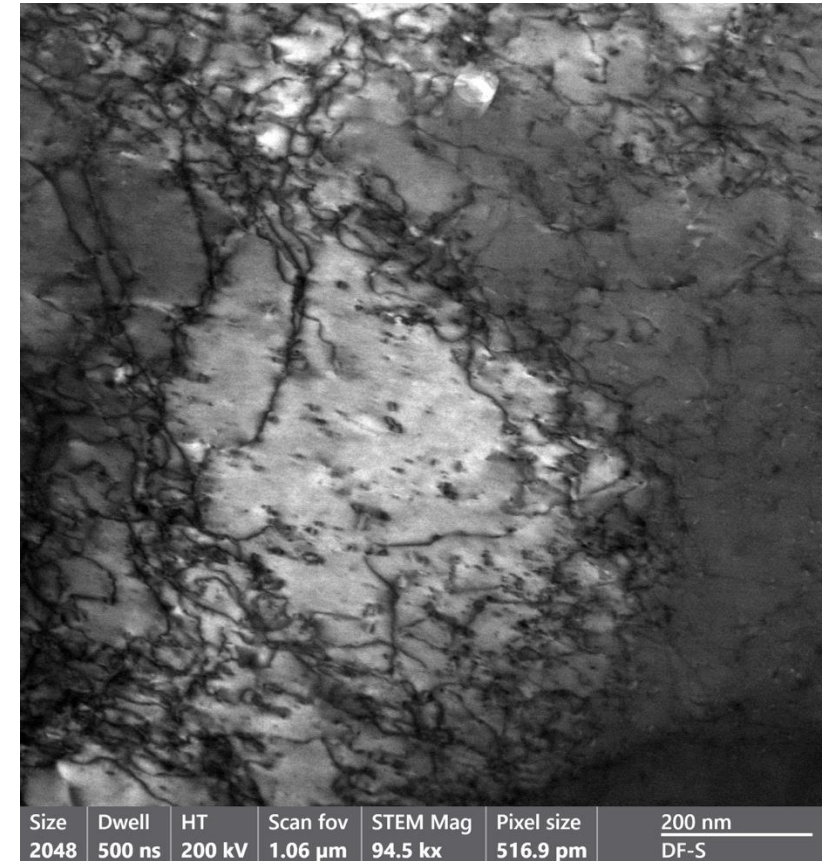
Lots of structures below the important region...



Bulk shows normal dislocation content. Similar to the structure observed by FIB without loop production



Size	Dwell	HT	Scan fov	STEM Mag	Pixel size	200 nm
2048	2.50 μ s	200 kV	1.06 μ m	94.5 kx	516.9 pm	HAADF

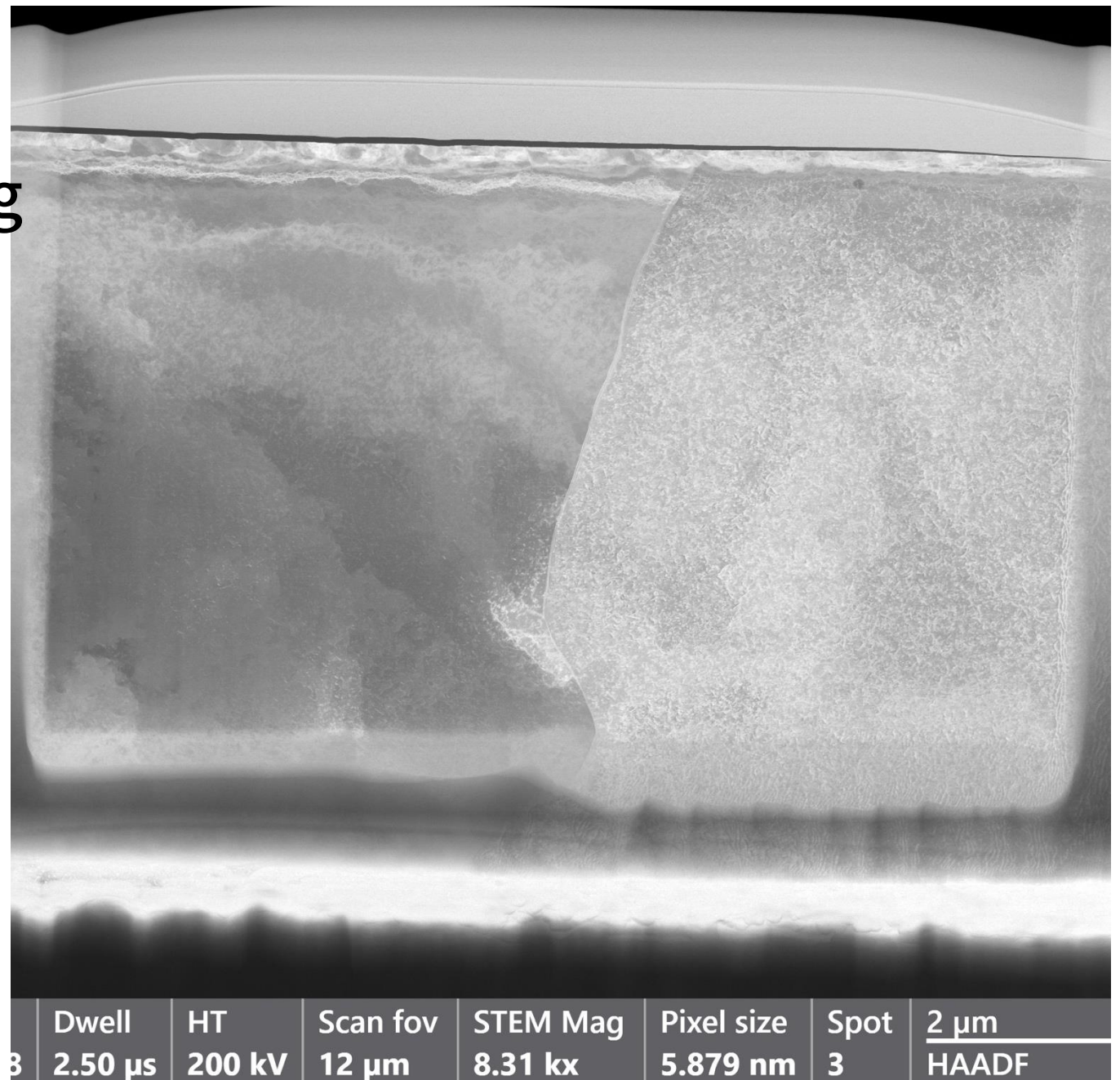


Size	Dwell	HT	Scan fov	STEM Mag	Pixel size	200 nm
2048	500 ns	200 kV	1.06 μ m	94.5 kx	516.9 pm	DF-S

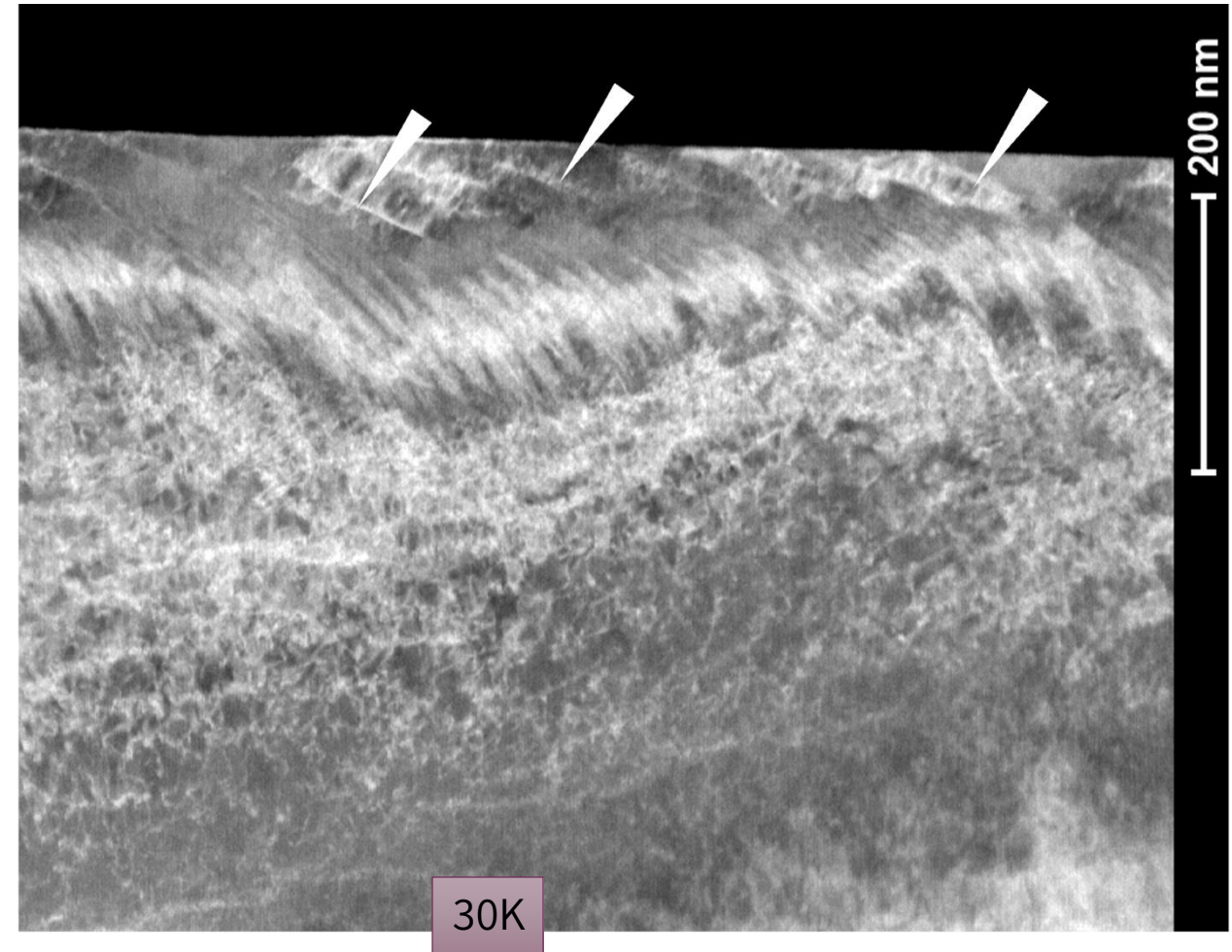
Size	Dwell	HT	Scan fov	STEM Mag	Pixel size	Spot	500 nm
2048	2.50 μ s	200 kV	2.99 μ m	33.4 kx	1.462 nm	3	HAADF

FIB of hard Cu limited but interesting

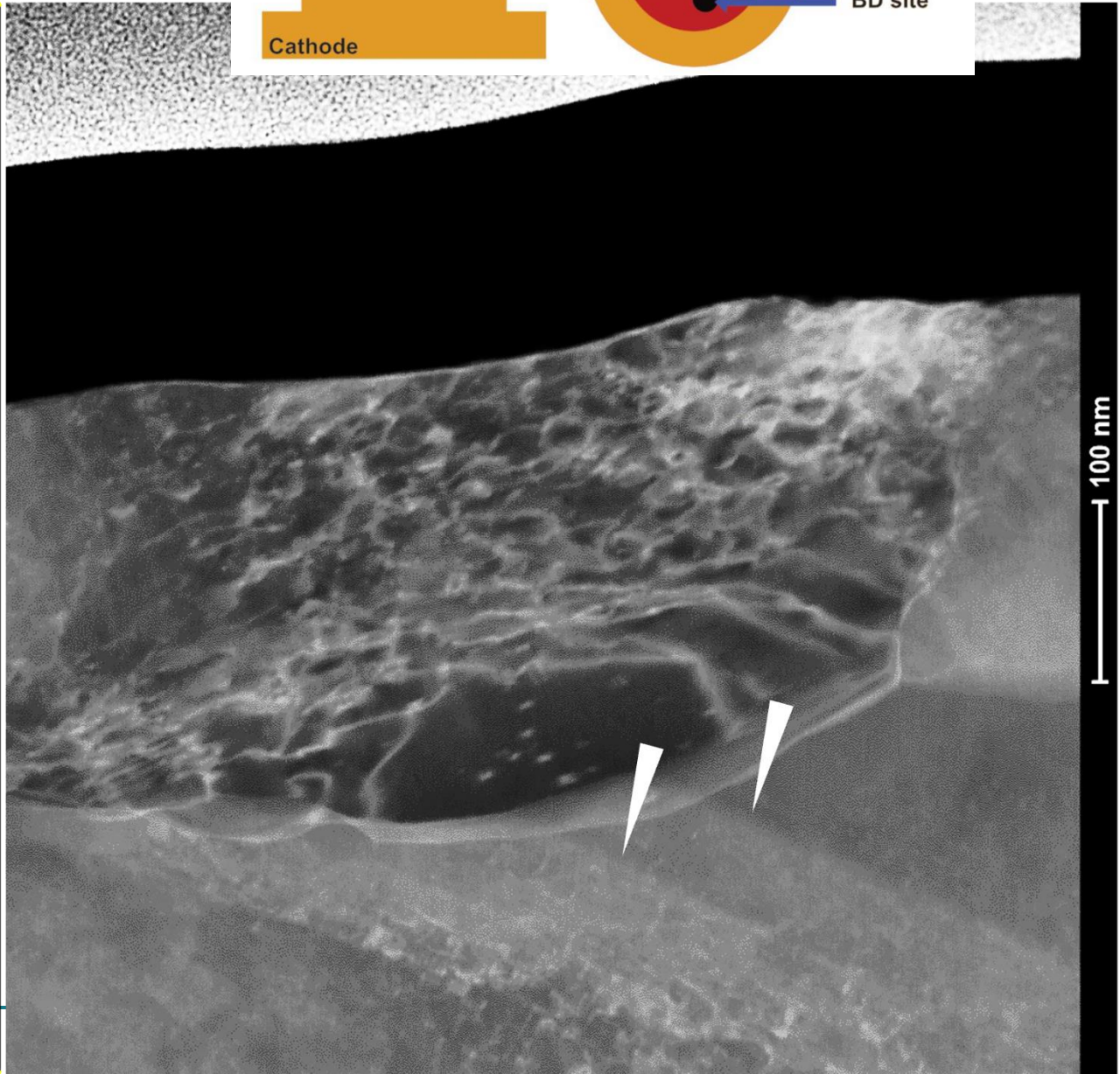
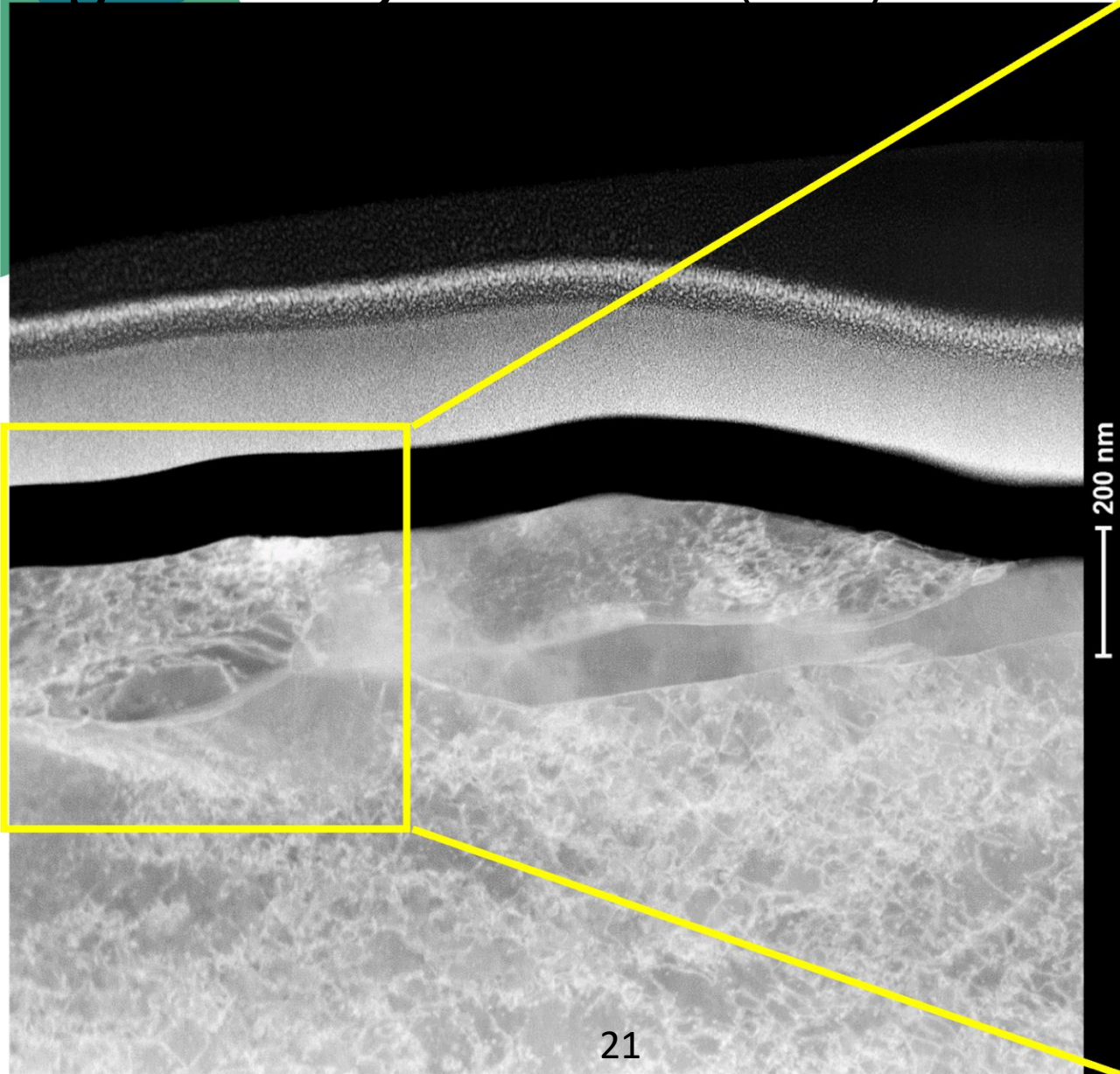
- Using cross-sectional FIB samples to study large-scale dislocation-related structures: grain boundaries, twins, etc, but not single loops or dislocations.
- Comparison of conditioned to unconditioned regions.



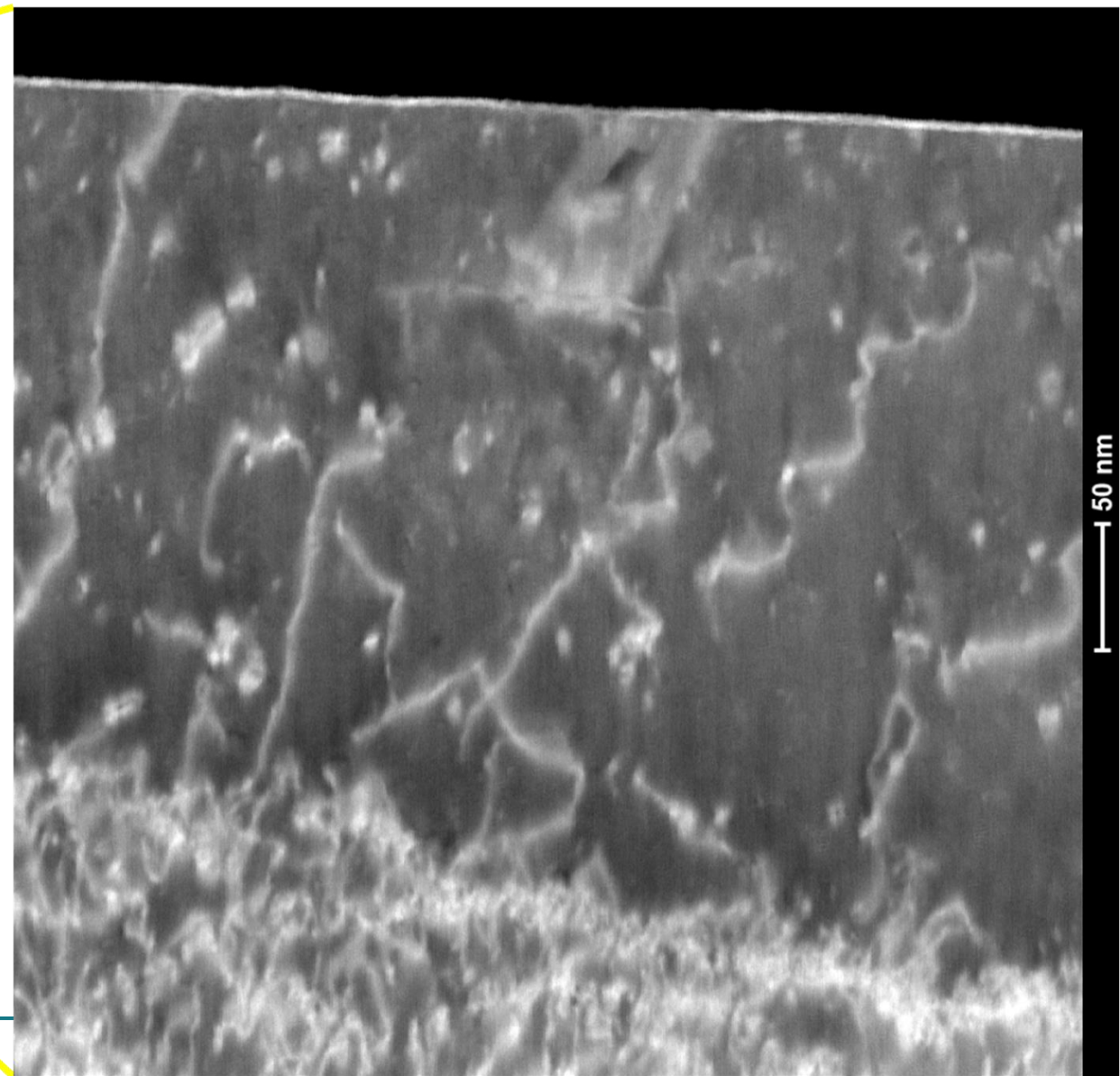
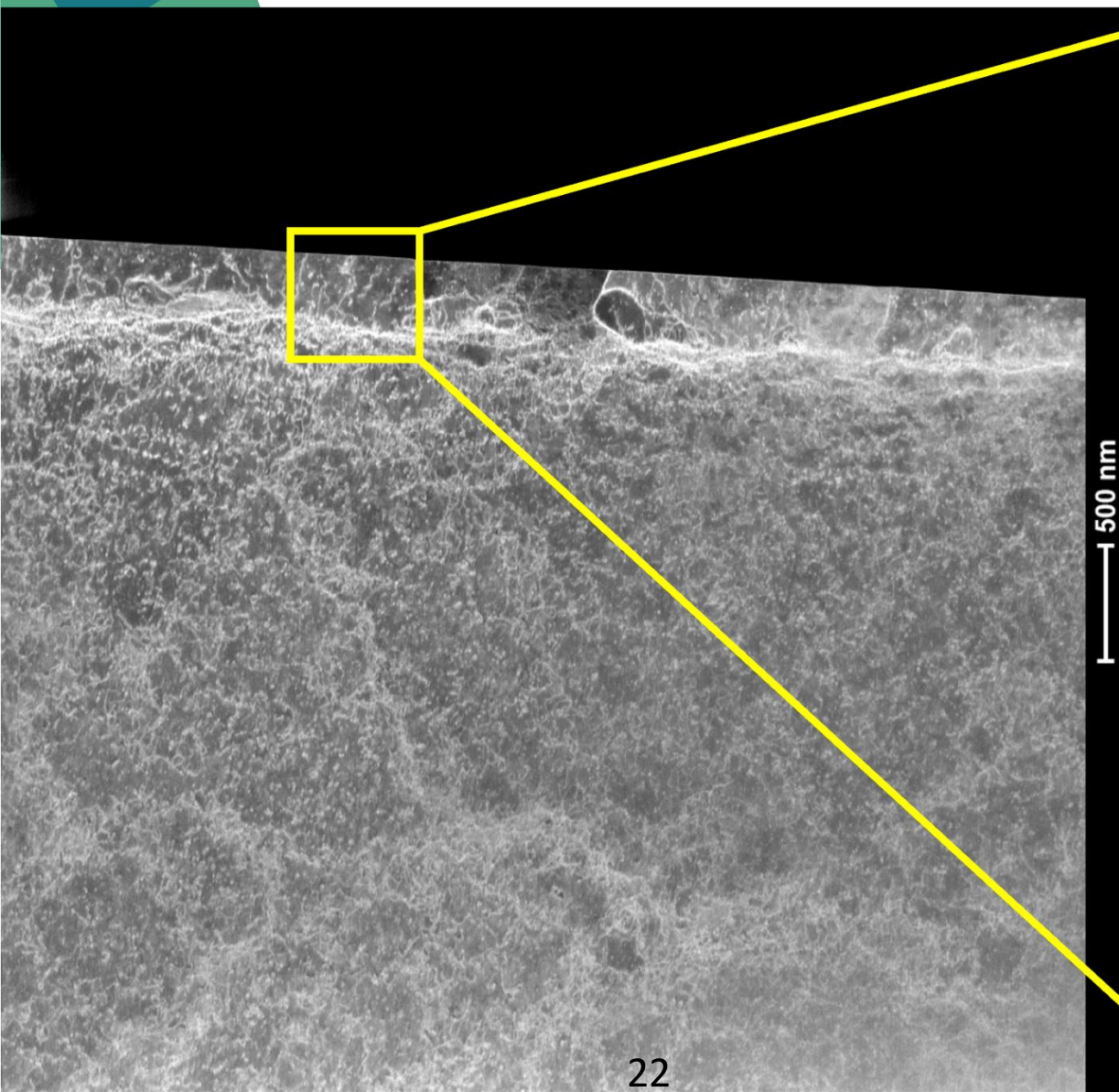
Cross-sectional STEM: Show regular, continuous structure up to the surface.

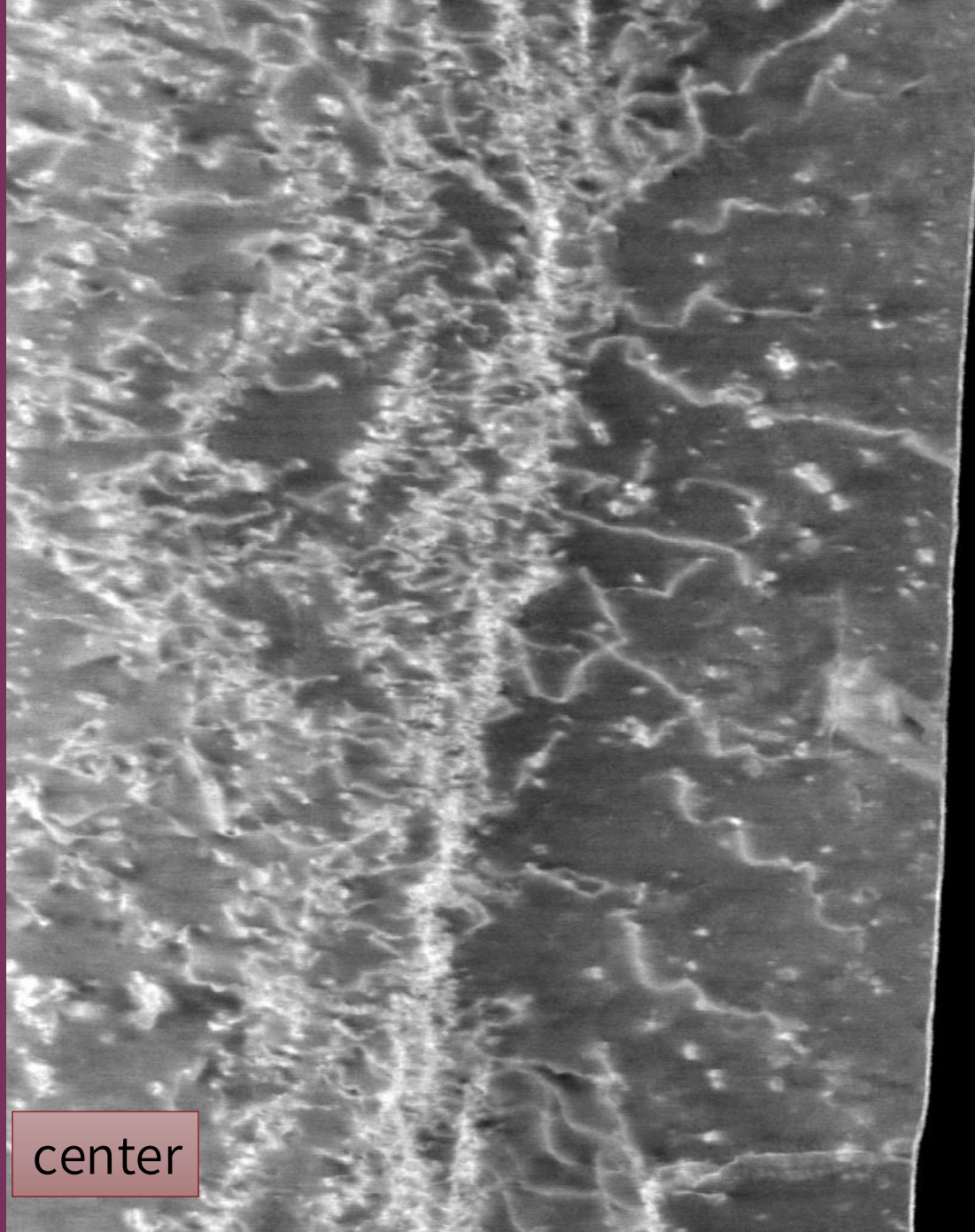
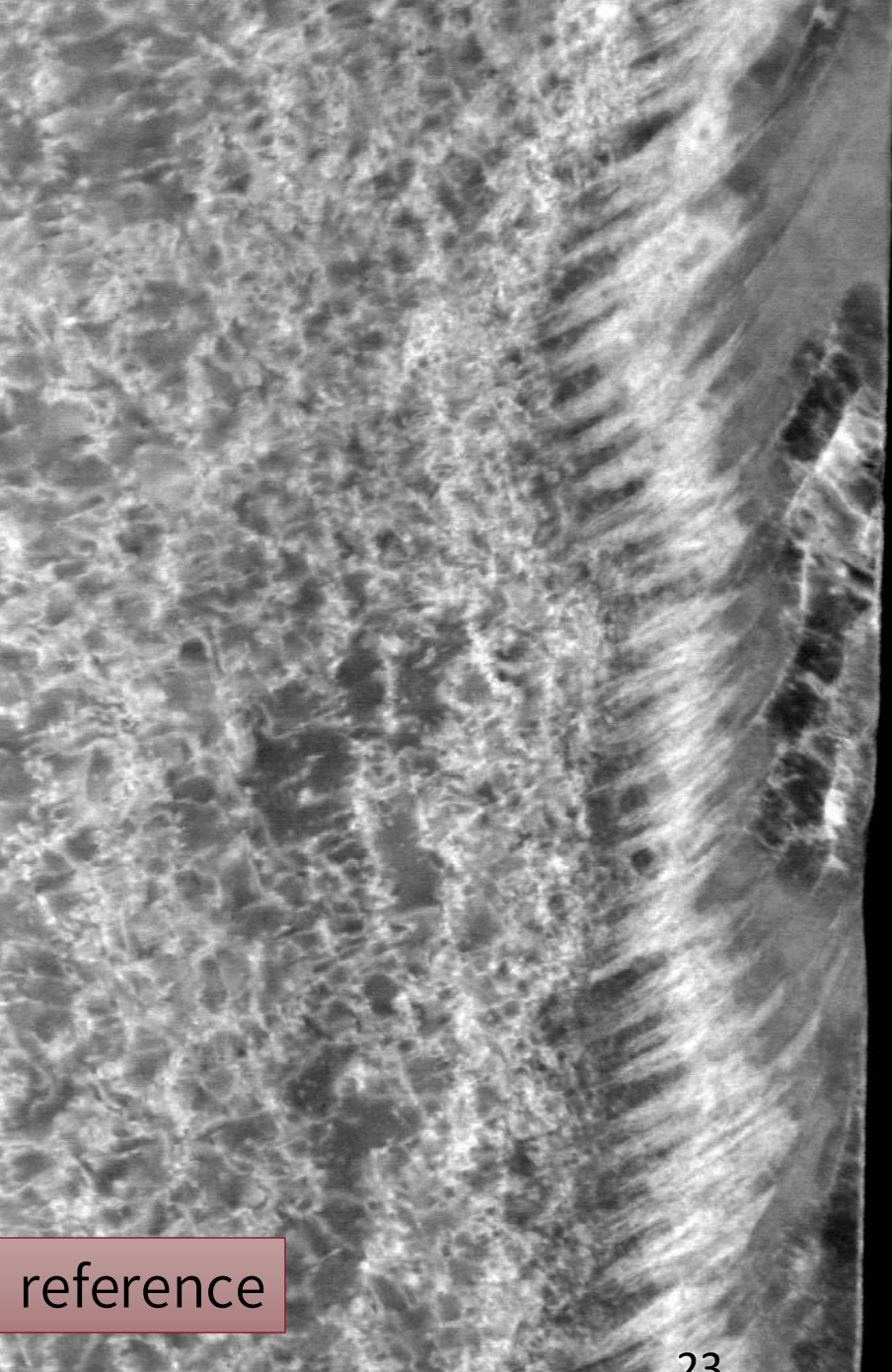


The top 100 nm of conditioned regions are significantly modified (300K)

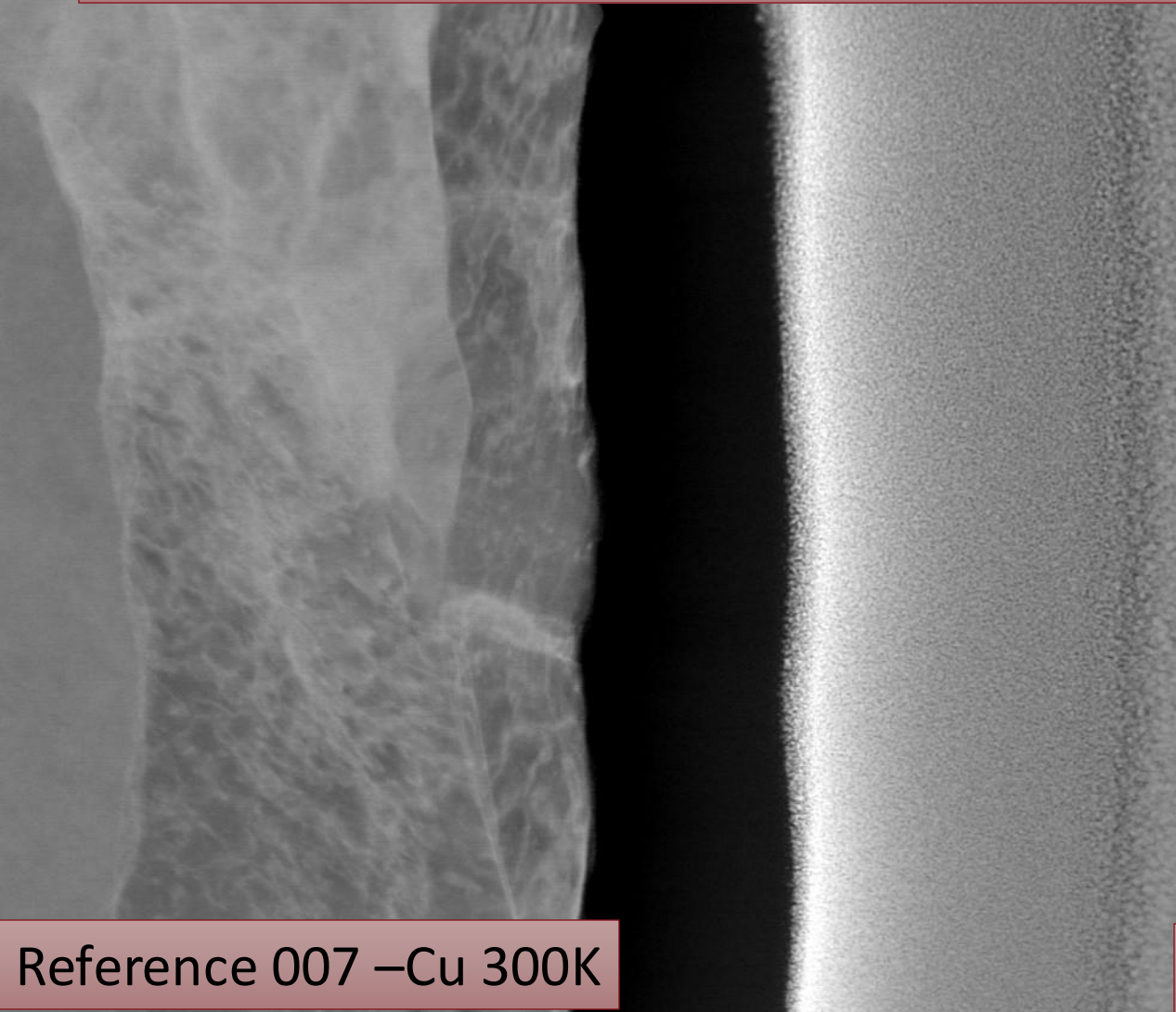


The top 100 nm of conditioned regions are significantly modified also @30K





Not all denuded zones are conditioned,
and not all “un-denuded” (dislocation rich) are not-conditioned



Reference 007 -Cu 300K

100 nm

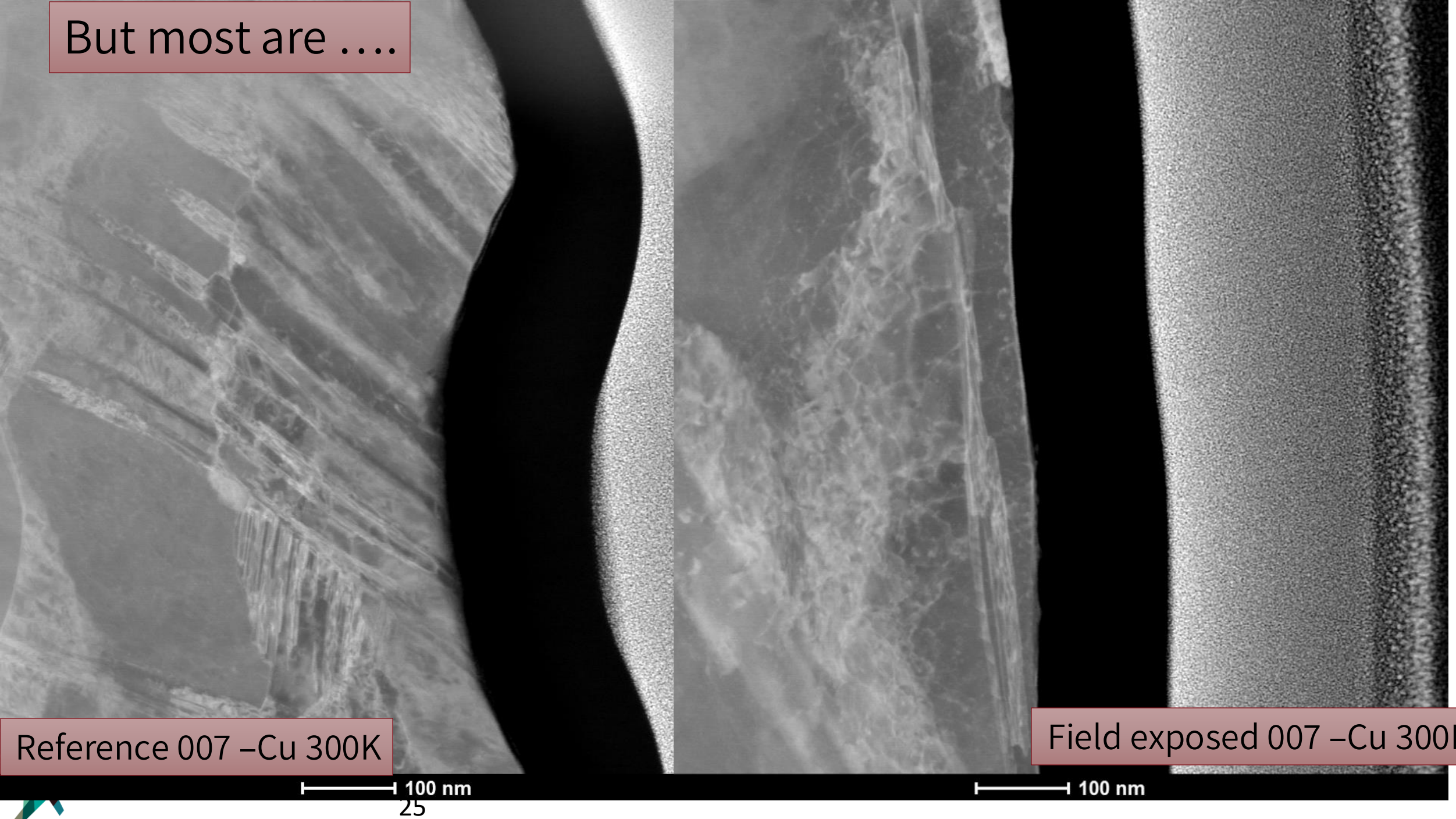
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Field exposed 007 -Cu 300K

100 nm

But most are

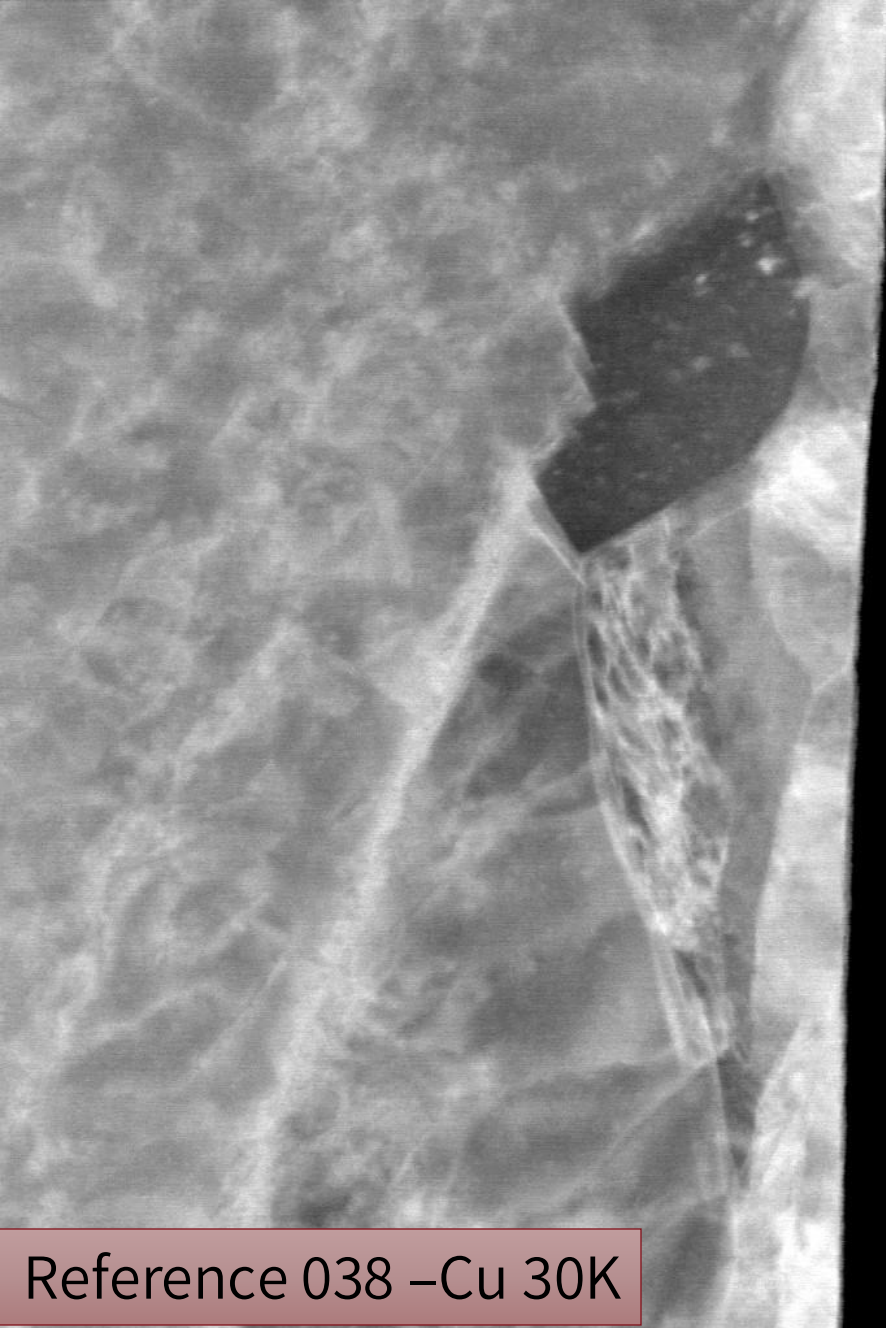


Reference 007 -Cu 300K

Field exposed 007 -Cu 300K

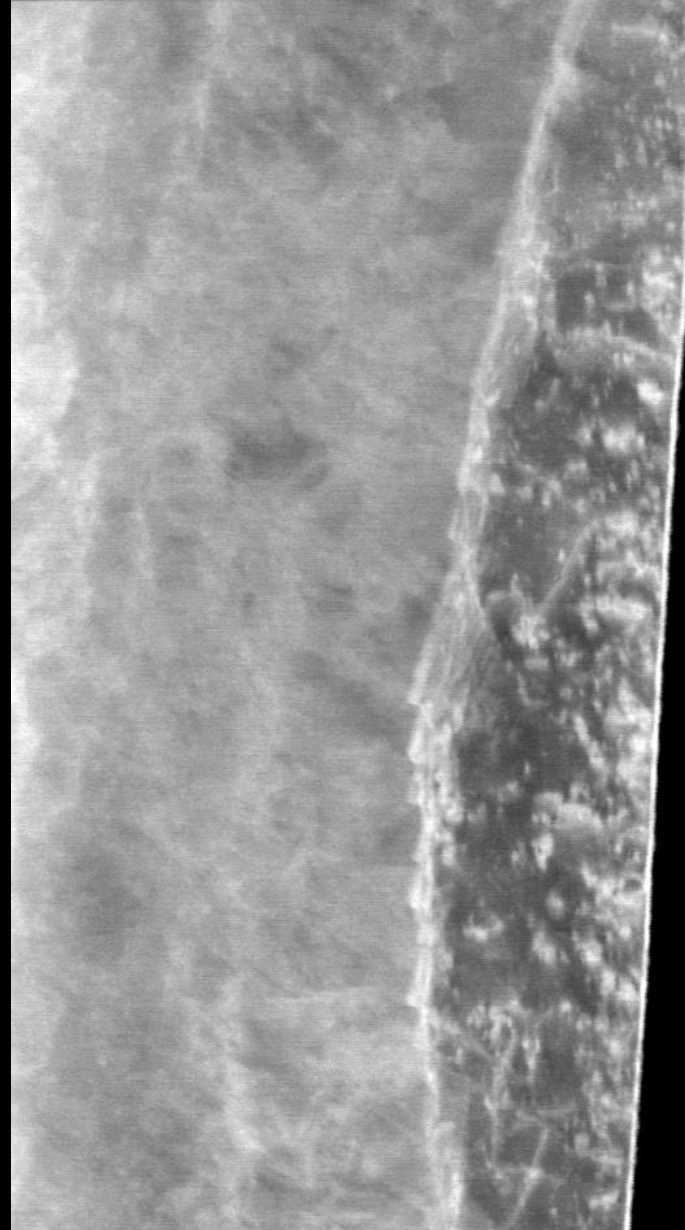
100 nm
25

100 nm



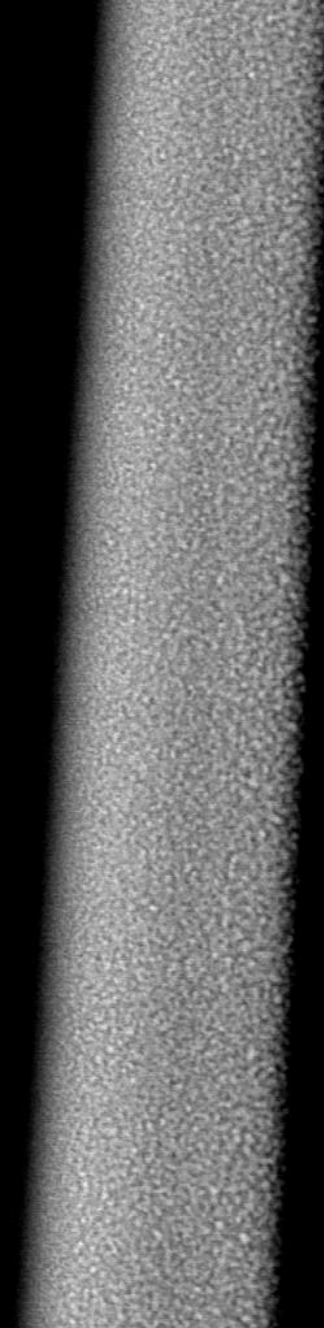
Reference 038 -Cu 30K

100 nm
26

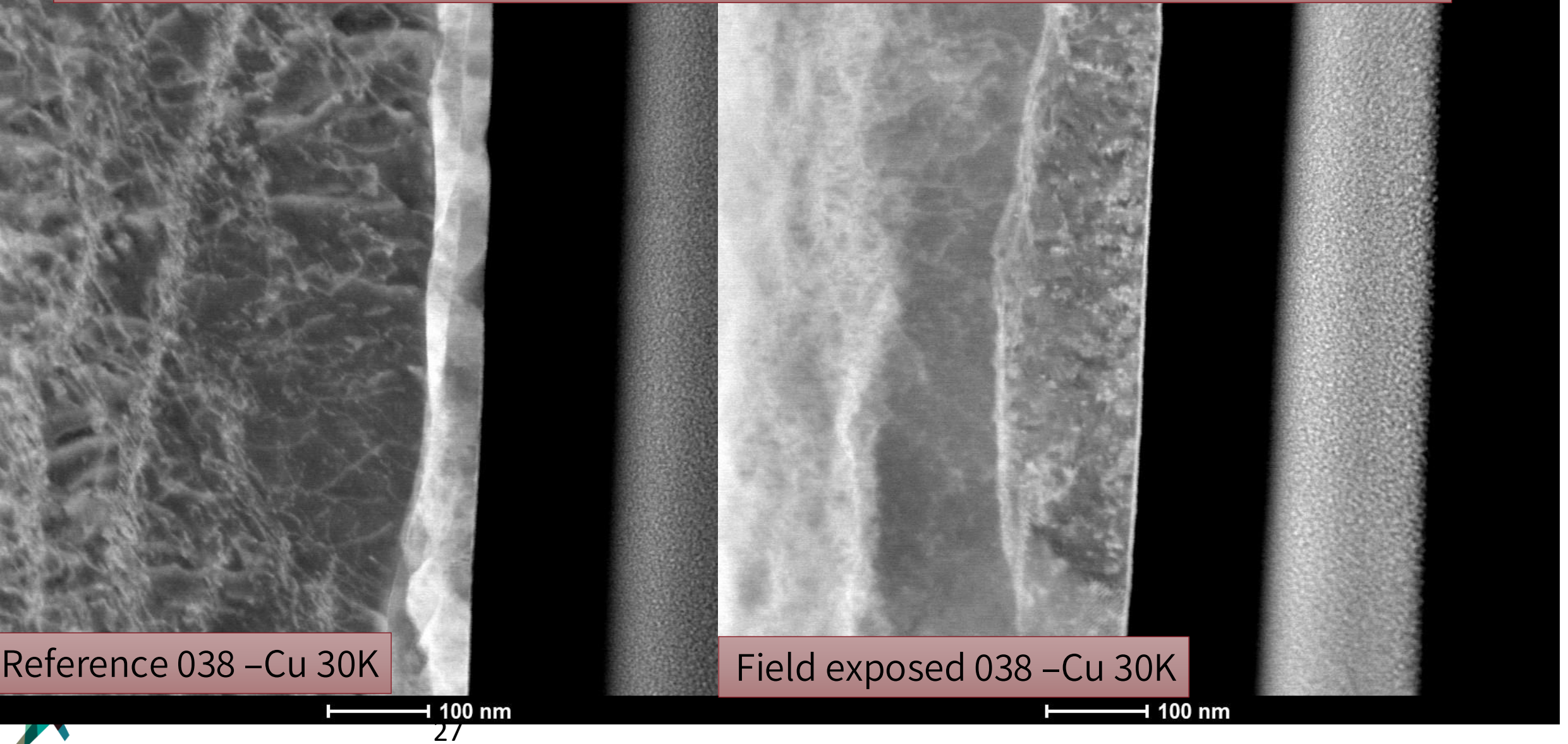


Field exposed 038 -Cu 30K

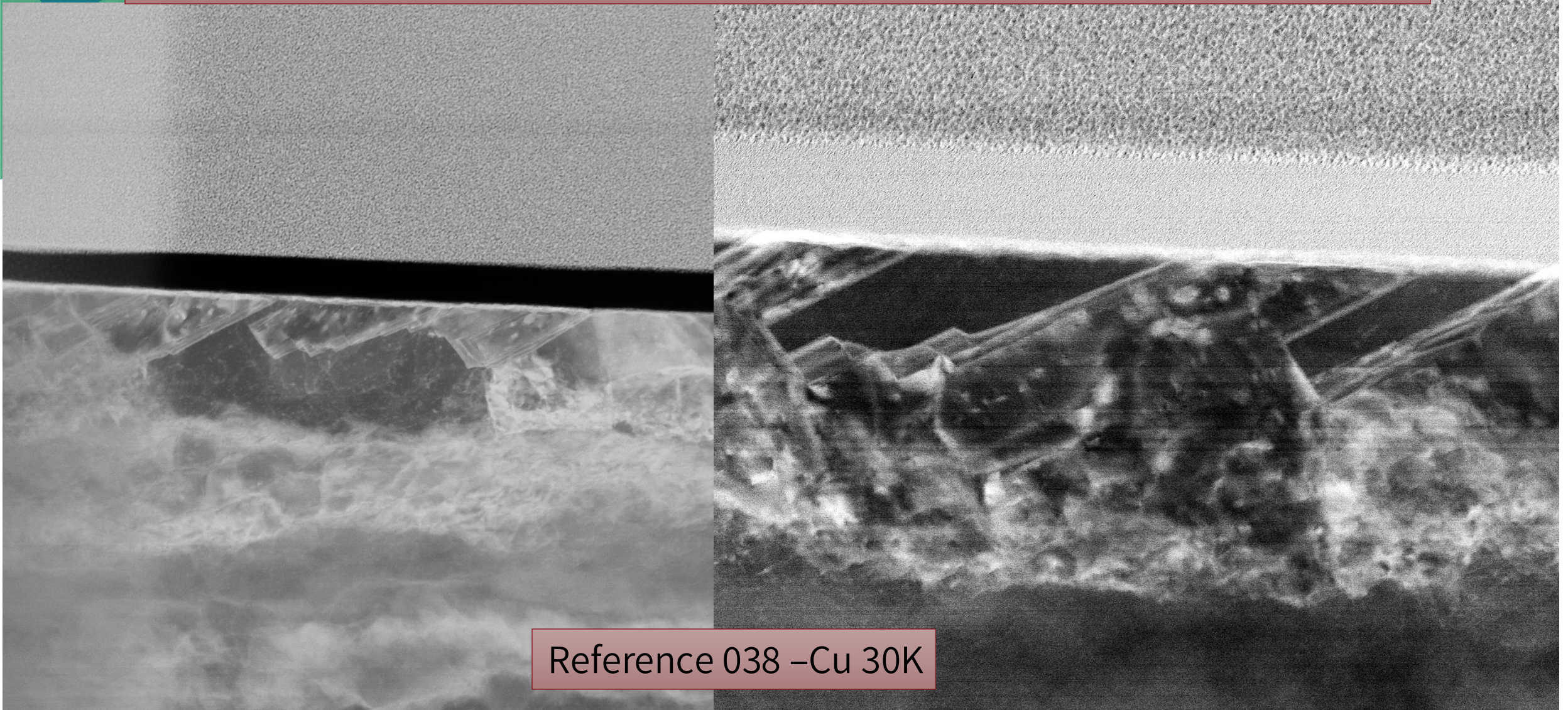
100 nm



Cryo-treated regions – have various types of grain refinement
but denuded zone formation is consistent



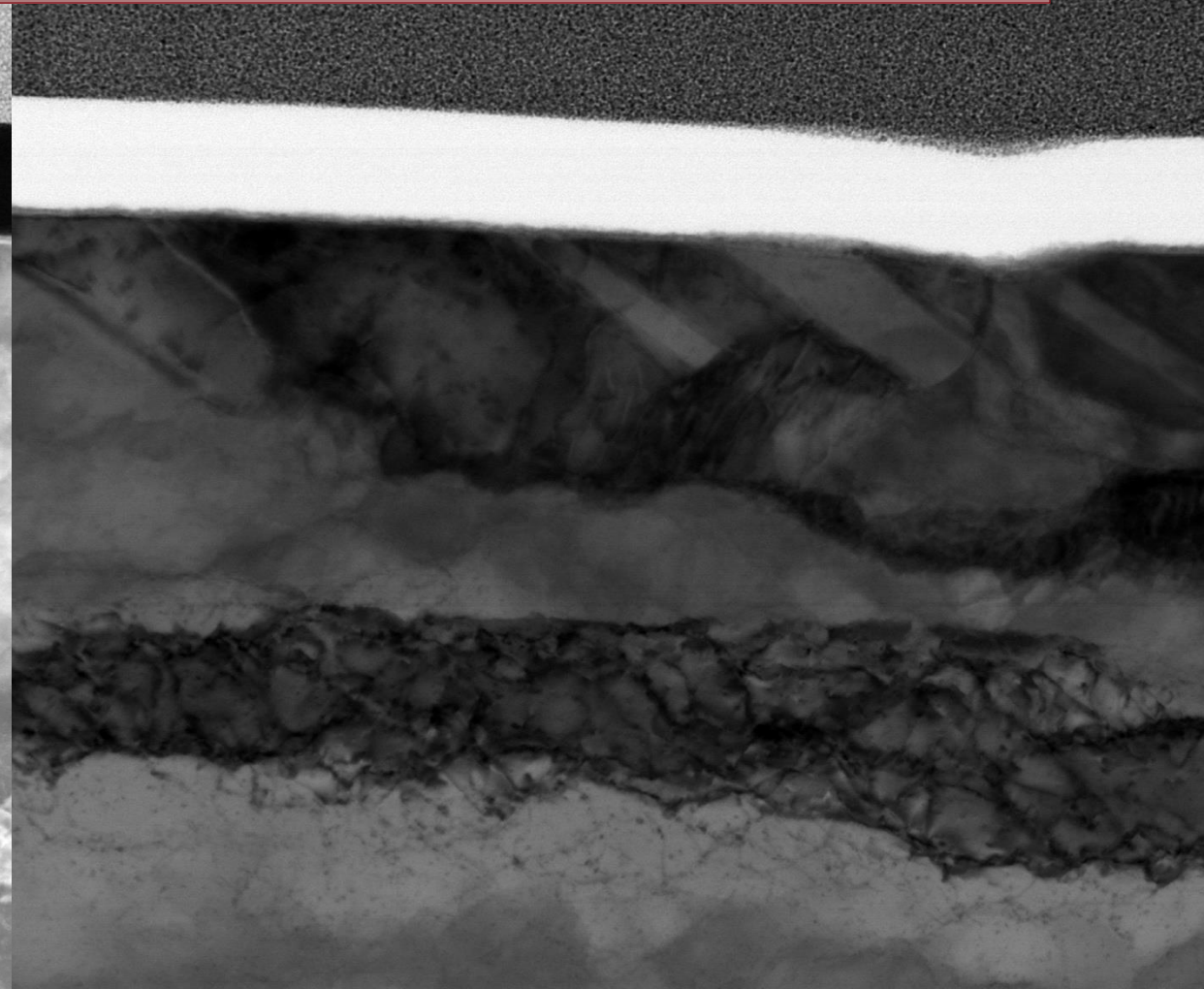
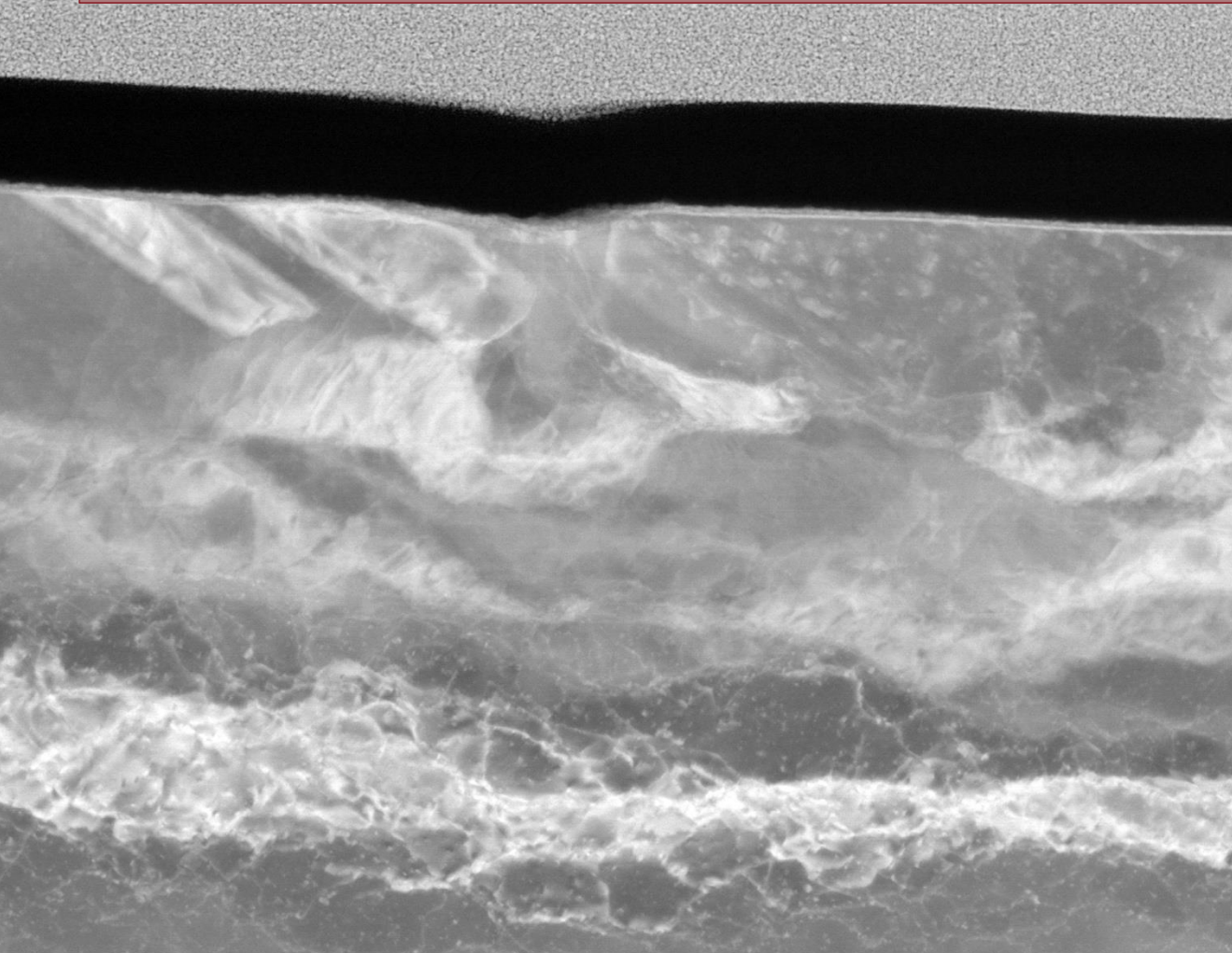
Sharp structures due to production,
Such features are not observed in the conditioned region



Reference 038 -Cu 30K

Size	Dwell	HT	Scan fov	STEM Mag	Pixel size	Spot	100 nm	Size	Dwell	HT	Scan fov	STEM Mag	Pixel size	Spot	50 nm
2048	2.50 μ s	200 kV	747 nm	134 kx	364.8 μ m	3	HAADF	2048	2.50 μ s	200 kV	374 nm	268 kx	182.4 μ m	3	HAADF

Field exposed – even when rare gbs are observed at the surface, consolidation of dislocations to gb. Arrays disappeared.



Field exposed 038 –Cu 30K

Size	Dwell	HT	Scan fov	STEM Mag	Pixel size	Spot	100 nm	Dwell	HT	Scan fov	STEM Mag	Pixel size	Spot	100 nm
2048	2.50 μ s	200 kV	747 nm	134 kx	364.8 pm	3	HAADF	2.50 μ s	200 kV	747 nm	134 kx	364.8 pm	3	DF-S

Overall – clear difference between conditioned and ref (Cu7@300K)



Summary and Outlook

Pursuing a direct link between the plastic mechanism and BD nucleation:

- A model offering a rare critical transition consistent with no observable localized emitter
- Consistent description of observed variation of BD rates.
- Consistent results for critical rare fluctuations in dark currents.

Microscopy

- It is a tricky business as sample manufacturing creates significant “red herrings.”
- However, larger-scale structures are not affected.
- Clear indications of significant structure evolution

Conditioned versus unconditioned regions:

- Established a method and its limits.
- Formation of denuded zones in hard Cu surfaces.
- The effect is limited to the top 100 nm.
- What next? Much more comparative forensics – Other materials? More samples?

